
Title: Distributed Electro-Hydraulic Systems and Drive Technologies for Construction Machinery: Principles, Architectures, Control, Challenges, and Future Perspectives

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Abstract: Traditional centralized hydraulic systems in construction machinery (CM) suffer from substantial throttling, pipeline, and load-mismatch losses, while prime-mover electrification alone cannot eliminate these system-level inefficiencies. Existing reviews have separately addressed electric and hybrid powertrains, energy regeneration, hydraulic architectures, and electro-hydraulic actuation, but an integrated synthesis of distributed system architectures (DSAs), drive units, control strategies, and engineering applicability is still lacking. This paper reviews distributed electro-hydraulic systems and related drive technologies for CM. The results show that DSAs form a technological spectrum from centralized power supply toward actuator-level independence rather than a universal replacement for centralized architectures. Electro-hydraulic actuators offer broader applicability for heavy-duty CM, whereas electromechanical actuators provide advantages in efficiency, precision, and leakage-free operation but require further validation in impact resistance and lifecycle reliability. Key challenges include communication, computation, fault tolerance, cybersecurity, and system-level validation. This paper can provide a reference for architecture selection, component development, and future research.

Keywords: Construction machinery; Electric drive technology; Distributed system architecture; Distributed drive unit; Distributed control; Energy efficiency.

1 Introduction

Construction machinery (CM), as core equipment in fields such as construction, mining, and water conservancy, has long adopted a hydraulically centralized power supply architecture in which a single power source drives multiple actuators. In such systems, hydraulic energy is delivered from the pump to individual actuators through pipelines and allocated by control valves, as illustrated in Fig. 1(a). This centralized power system, which relies on open-loop throttling control, inherently exhibits low overall system energy efficiency (SEE). The inefficiency arises from complex internal energy transmission chains, extended power transfer paths, and pronounced load differences among multiple actuators [Yang et al., 2022]. As a result, throttling losses and pipeline losses are generally high, leading to poor system-level efficiency [Fassbender et al., 2021] and making these systems one of the major contributors to greenhouse gas emissions [Helms and Heidt, 2014]. According to studies conducted by the National Fluid Power Association of the United States, the average SEE of fluid power systems across various industries is only about 22%, with the minimum dropping to as low as 9%

[Love et al., 2012]. Existing studies further indicate that the overall SEE of CM driven by diesel-hydraulic systems is typically only 15-30% [Ochiai et al., 2008; Ge et al., 2019; Excess Engineering, n.d.]. In recent years, although advances in electric-drive technologies have improved the overall efficiency of pure-electric construction machinery (P-ECM) to 30-45% [Hao et al., 2023; Podgornyy et al., 2024], this still implies that more than half of the electrical energy is wasted in each operation, despite the battery accounting for nearly half of the total machine cost. Compared with mechanical transmission systems, which can achieve energy utilization efficiencies of approximately 90%, this contrast further highlights the efficiency bottleneck inherent in centralized hydraulic drive systems [Ge et al., 2019]. Meanwhile, owing to its advantages in high power density, strong adaptability to impact and variable loads, and cost effectiveness [Kazama et al., 2019], hydraulic technology remains difficult to replace with alternative technologies in the CM sector in the near term, rendering the improvement of hydraulic SEE an increasingly urgent issue.

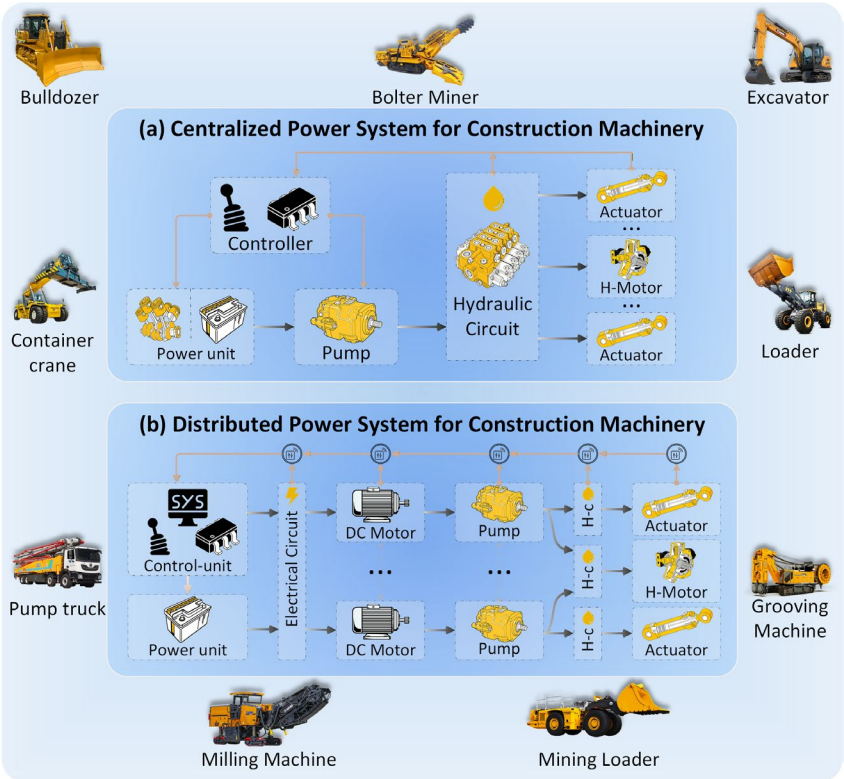


Fig. 1 Schematic comparison of CM system architectures: (a) conventional CSA with centralized hydraulic power supply and (b) DSA with actuator-level independent power supply.

Table 1 List of Abbreviations.

Abbreviation	Full term	Abbreviation	Full term
CM	Construction Machinery	ICE	Internal Combustion Engine
CSA	Centralized System Architecture	IoT	Internet of Things
DSA	Distributed System Architecture	LS	Load-Sensing
EDT	Electric Drive Technology	MPC	Model Predictive Control
EHA	Electro-Hydraulic Actuator	P-ECM	Pure-Electric Construction Machinery
EMA	Electromechanical Actuator	SEE	System Energy Efficiency

Against this backdrop, distributed system architectures (DSA), owing to their high SEE and flexibility, are increasingly regarded as a potential alternative to centralized system architectures (CSA) [Wang et al., 2023b]. The distributed concept originated in the field of computer science and refers to systems composed of multiple mutually independent nodes that operate collaboratively through a network to accomplish common tasks [Díaz et al., 2024]. This architecture emphasizes modular functional decomposition and coordinated optimization, leveraging efficient communication mechanisms and advanced control algorithms to achieve high performance and reliability. In the aerospace sector, distributed technologies—such as electromechanical actuators (EMA) and electro-hydraulic actuators (EHA)—employ multiple independent drive modules acting on critical components, thereby significantly enhancing SEE, fault tolerance, and adaptability [Böhnisch et al., 2026]. Benefiting from the multi-port parallel power supply characteristics of batteries, DSA exhibit distinct advantages in the field of CM, as illustrated in Fig. 1(b). In this configuration, each actuator is equipped with its own prime mover and transmission unit, enabling independent power supply through decentralized energy sources. This arrangement effectively mitigates the influence of load disparities among multiple terminals on actuator operation, significantly shortens power supply paths, and reduces energy losses during power distribution. Consequently, DSA feature system decoupling, high operational SEE, fast dynamic response, and strong fault isolation capability [Inderelst et al., 2020]. By reducing the energy losses inherent in centralized power supply, DSA offer a potential route to improving energy efficiency and performance under variable-load, high-impact, and strongly coupled CM operating conditions.

To address the challenges of energy consumption and efficiency in CM, extensive research efforts have been undertaken by both academia and industry across several key directions, including drive modes, such as fully electric systems [Fassbender et al., 2021; Huang et al., 2024; Opgenoorth et al., 2020] and hybrid drives [Zhang et al., 2024; He et al., 2018], system architectures [Huang et al., 2024; Quan et al., 2021; Padovani et al., 2020], operating condition characteristics [Dadhich et al., 2016; Wang et al., 2017; Borngund et al., 2024], energy management [Wang et al., 2017; Trinh et al., 2022; Wang et al., 2016], energy regeneration [He et al., 2020; Do et al., 2021], and intelligent technologies [Truong et al., 2018; Zhe et al., 2023; Ding et al., 2018; You et al., 2023; Eraliev et al., 2022]. Significant progress has been achieved in these areas. In particular, P-ECM has demonstrated strong development potential by enabling zero-emission operation, while improvements in cost competitiveness and overall machine performance have supported growing market acceptance [China Daily, 2024]. The expanding body of research and review literature on the SEE and performance of CM further highlights the importance of improving energy efficiency at both the power-source and system levels.

At present, most studies on emerging technologies represented by electric drives focus on replacing traditional internal combustion engines (ICE), while the front-end working systems largely retain hydraulically driven CSA originally designed for ICE. This continuity constrains the further exploitation of the potential of EDT, making DSA and distributed power-supply modes an important route for overcoming existing SEE bottlenecks. Fassbender et al. proposed a comparative framework of “distributed independent drive versus centralized drive,” discussed the respective advantages of the two modes, and identified EHA as a key pathway toward the electrification of hydraulic tools [Fassbender et al., 2021]. Inderelst et al. conducted similar studies and further analysed the potential of substituting hydraulic power with electrical energy [Inderelst et al., 2020]. Although numerous studies have explored distributed concepts and related technologies in CM, the field still lacks a comprehensive synthesis linking system architectures, distributed drive units, control strategies, and engineering applications. Consequently, understanding of DSA remains fragmented across academia and industry, underscoring the need for systematic review and integration.

Based on this background, this review examines the development of distributed concepts and technologies for CM, with particular emphasis on DSA, distributed drive

units, and distributed control. The review traces the formation and evolution of different technological pathways and analyses their application characteristics, applicability, and engineering challenges. Relevant studies and application cases from journals, conference proceedings, English-, Chinese-, and German-language publications, and industry sources are integrated to establish a technological framework from the perspectives of system architecture, drive technology, and control methodology. The resulting synthesis is intended to provide a reference for architecture selection, component development, and future research on DSA for CM.

To improve the transparency and reproducibility of the literature retrieval and screening process, multiple databases and search engines were used to identify and synthesise relevant published or publicly available academic literature and industry materials, following literature-review methodologies widely recognised by the research community [Zhao *et al.*, 2022]. The specific retrieval and screening procedures are described below.

- (1) **Information Sources:** Academic literature was primarily retrieved from Web of Science, Google Scholar, and the China National Knowledge Infrastructure (CNKI), with an emphasis on peer-reviewed journal articles and conference papers. Given the strong engineering and application-oriented nature of distributed technologies for CM, supplementary searches were also conducted through Google and the official websites of relevant companies and research institutions to identify representative industry news, product releases, and engineering application cases. This approach enabled the review to capture technological developments from both academic research and industrial practice.
- (2) **Search Keywords and Combinations:** Based on the scope of this review and the principal technical themes represented in the included literature, the search terms were grouped into four categories. The first category comprised keywords related to the research objects, including “construction machinery,” “construction equipment,” “mobile machinery,” “excavator,” and “loader.” The second category covered system architectures, including “distributed system,” “distributed architecture,” “decentralized system,” “distributed power,” and “independent drive.” The third category focused on drive technologies, including “electro-hydraulic actuator,” “EHA,” “electromechanical actuator,” “EMA,” “pump-controlled actuator,” and “electro-hydraulic drive.” The fourth category addressed control and energy management, including “distributed control,” “cooperative control,” “energy management,” “power sharing,” “energy regeneration,” and “fault-tolerant control.” Corresponding Chinese keywords and their combinations were used for searches in Chinese-language databases.

During the literature search, keywords describing the research objects were cross-combined with those related to system architectures, drive technologies, and control and energy management, rather than being used individually. For example, a representative English search string was: “construction machinery” AND (“distributed system” OR “distributed architecture” OR “decentralized system”). In addition, the search terms were iteratively supplemented according to frequently occurring terminology identified during the initial search, keywords used in representative publications, and alternative technical terms adopted by different research groups, thereby reducing the risk of omitting relevant studies because of terminological variations.

- (3) **Literature Screening Scope and Procedure:** The literature coverage period was limited to January 2010–31 August 2026 to cover the major stages of development since distributed concepts began to be introduced into CM research. Duplicate records were first removed, after which the remaining publications were screened sequentially by title and abstract and then subjected to full-text review. Studies were included if they met at least one of the following criteria: (i) investigation of distributed or partially distributed power-supply architectures for the working systems of CM; (ii) investigation of independent electro-hydraulic or electromechanical drive units for

CM, such as EHA, EMA, and related pump-controlled actuators; (iii) investigation of distributed control, cooperative control, energy management, or fault-tolerant control in multi-actuator systems; or (iv) representative industry sources reporting progress in the engineering implementation of relevant technologies. Studies that used the term “distributed” only in a general sense, without direct relevance to power transmission, actuator drive, or system control in CM, as well as duplicate publications and sources lacking sufficient technical information, were excluded from the statistical analysis.

- (4) Evidence Sources and Statistical Scope:** Peer-reviewed journal articles and conference papers were given priority as evidence when synthesizing technical performance, system characteristics, and research findings. For topics for which formal academic publications were not yet available but which reflected progress in industrial prototypes, product development, or commercialization, publicly verifiable company materials, industry news, and information released by research institutions were used as supplementary evidence. Based on the above search and screening procedures, a total of 132 academic publications published between January 2010 and 31 August 2026 were included in the bibliometric dataset, comprising 102 journal articles and 30 conference papers. In addition, 10 representative industry news reports and online sources were used as supplementary evidence. These materials constituted the primary dataset for the subsequent bibliometric analysis and technical review.

It should be noted that, because distributed technologies for CM span both academic research and industrial development, the bibliometric results of this review are subject to several limitations. First, databases such as Web of Science and CNKI differ in their coverage of publication sources, languages, and geographical regions. Consequently, some grey literature, corporate technical reports, patent documents, and non-English industrial materials may not be fully indexed. Although Google Scholar and general web searches broaden the scope of information retrieval, their coverage and ranking mechanisms cannot be fully standardized. Second, conference papers constitute a non-negligible proportion of the academic literature in this field, and differences in database coverage and publication cycles across publication types may affect annual publication counts and the distribution of publication sources. In addition, relevance screening inevitably involves author judgement in defining the boundaries of the research field. Accordingly, the bibliometric results presented here are more appropriately interpreted as indicators of overall development trends, major technological pathways, and representative research groups rather than as an exhaustive enumeration of all relevant research outputs worldwide. Factors such as database coverage, document language, and screening criteria may affect the absolute numbers and relative proportions associated with different years, countries, institutions, and publication sources; however, they do not alter the primary objective of this review, which is to trace the major technological evolution pathways of distributed technologies for CM.

Based on the above dataset, this review further analyses the development trends of distributed technologies for CM in terms of publication year, publication source, and representative research institutions. As shown in Fig. 2, the publication output in this field exhibits two relatively distinct peaks, around 2018 and 2024. The increase around 2018 was closely associated with the gradual introduction of distributed power supply, independent pump control, and EHA technologies into CM. After 2020, with the acceleration of electrification and the continued maturation of electric drive, power electronics, and modular actuator technologies, the advantages of DSA in system decoupling, independent power supply, and energy management attracted increasing attention, which may have collectively contributed to the growth in related publications around 2024 [Fassbender et al., 2021]. In addition, advances in digital technologies, including sensing, industrial communication, the Internet of Things (IoT), and edge computing, have provided important technical support for state sensing, information exchange, and coordinated control in multi-node systems and may also have indirectly contributed to the growing research interest in DSA [Wang et al., 2024; Xu et al., 2023].

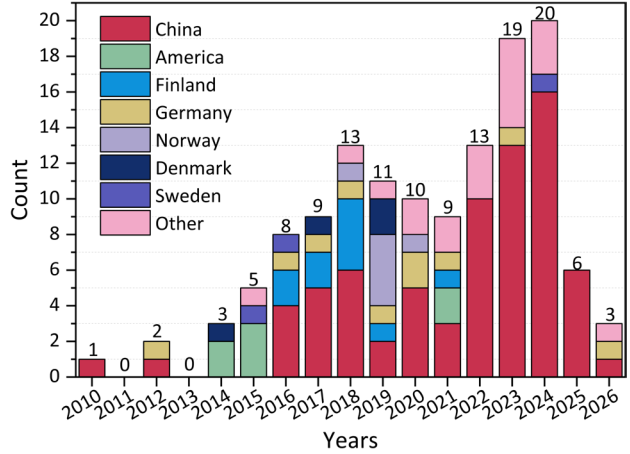


Fig. 2 Distribution of academic publications by publication year and country.

In terms of publication sources, relevant studies are mainly concentrated in journals such as *Automation in Construction*, *Energy*, and *Energy Conversion and Management*, as well as fluid-power conferences such as FPMC, as summarized in Table 2. Conference papers account for approximately 22.7% of the academic publications included in the bibliometric dataset, indicating that the field remains characterized by architectural exploration, prototype validation, and rapid technological iteration, for which specialized conferences provide an important channel for communicating new system configurations, drive concepts, and control methods. This also suggests that relying solely on journal articles may overlook some technical developments that remain at an early validation stage. Current research mainly focuses on DSA for CM, drive-unit design, energy management, coordinated control, and comparisons of the applicability of different system architectures. Among various types of CM, excavators remain one of the most intensively studied and validated platforms for distributed technologies because of their high installed hydraulic power, substantial load variations, and the large proportion of total energy consumption attributable to the working equipment.

Table 2 Major publication sources in the bibliometric dataset.

Rank	Publication	Count
①	Automation in Construction	13
②	Energy	7
③	FPMC (Conference proceedings)	6
④	Energies	5
⑤	Energy Conversion and Management	4
★	Other conference proceedings	24

Figure 3 further presents the representative research institutions identified from the literature search. The institutions shown in the figure cover the majority of the representative research outputs included in the dataset. Based on the available sample, representative studies in Europe and China are more frequently conducted by universities and research institutes, whereas industry participation is relatively more prominent among representative studies from the United States. These regional differences in the composition of research organizations reflect, to some extent, different emphases in research orientation, technological maturity, and pathways for technology transfer. University-led research generally places greater emphasis on system architectures, fundamental principles of key components, control methods, and proof-of-concept validation, whereas studies with stronger industrial participation tend to focus more on

prototype integration, engineering validation, and product-oriented applications. Accordingly, differences in institutional composition can, to some extent, indicate the relative emphasis placed by different research actors on fundamental research versus engineering translation.

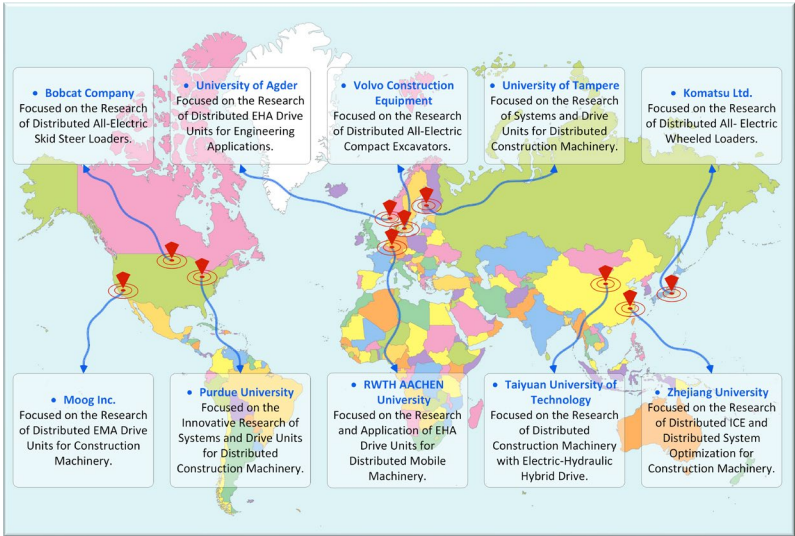


Fig. 3 Research institutions of DSA in CM with representative achievements.

The structure of this paper is illustrated in Fig. 4. Section 2 introduces the concepts, characteristics, and classification of DSA in CM. Section 3 reviews representative DSA architectures and their technological evolution. Section 4 summarizes the characteristics, applicability, and research progress of EHA and EMA. Section 5 reviews distributed control methods and identifies key research gaps. Section 6 discusses engineering challenges and application adaptability. Finally, Section 7 summarizes the main conclusions and outlines future research directions.

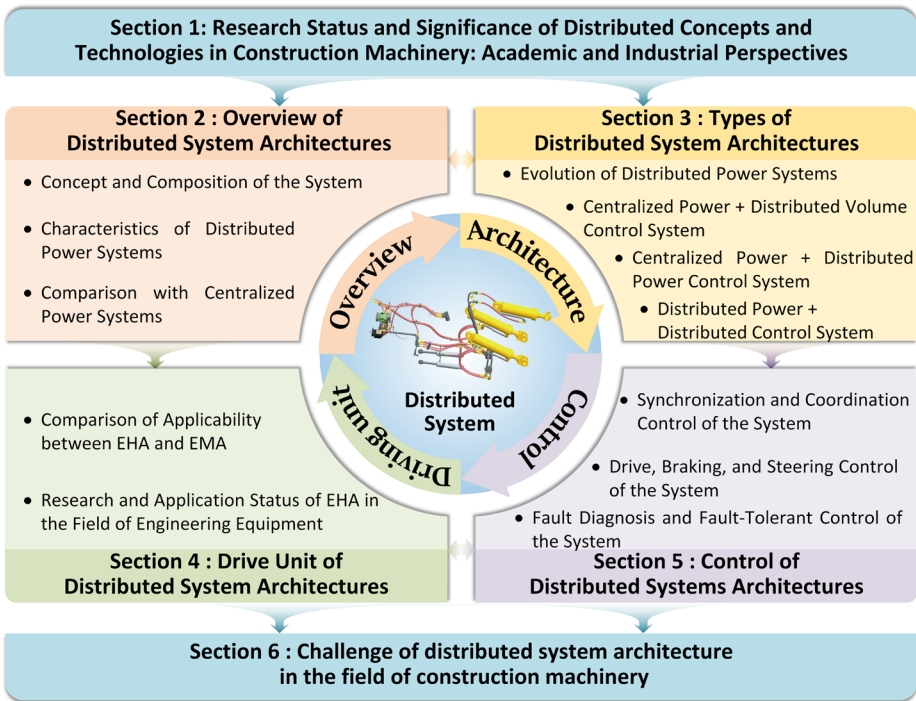


Fig. 4 Structure of this review paper.

2 Overview of DSA

This section provides a systematic overview of DSA in CM, integrating current research findings and practical applications of distributed concepts and technologies in the field.

2.1 Concepts and Components

2.1.1 Concept Definition

A DSA for CM refers to a system configuration in which multiple actuators are each equipped with decentralized power units to provide independent power supply and energy transfer. Leveraging modular design, intelligent control, and efficient communication mechanisms, the system enables distributed monitoring and diagnostics at the system level. The core concept lies in overcoming the limitations of traditional CSA, which rely on a single power source and long, complex energy transmission paths, by embedding power sources and control units at the actuator level. This allows for independent energy supply and precise regulation, providing efficient and flexible power support for complex operating conditions, as illustrated in Fig. 5. Within DSA, multiple power sources—such as batteries, electric motors, or electro-hybrid systems—and their energy transmission pathways are coordinated through power-sharing mechanisms, ensuring that each actuator can flexibly adjust its energy supply under variable load conditions [Wang et al., 2023b]. This coordination improves energy allocation and reduces unnecessary power losses, thereby enhancing overall SEE.

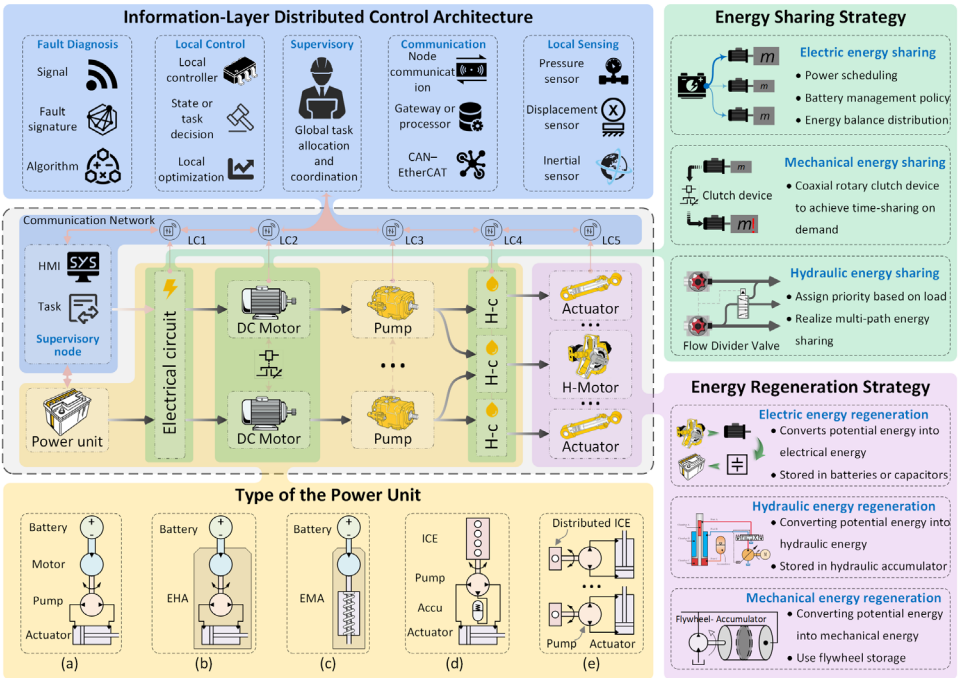


Fig. 5 Overview of DSA in CM, illustrating the coordination between the information-control layer and physical energy-flow layer, together with five representative power-unit configurations and the corresponding electrical, mechanical, and hydraulic energy-sharing and regeneration mechanisms.

2.1.2 System Composition

The DSA for CM primarily comprises four components: the power unit, actuator unit, control and coordination unit, and energy management unit, detailed as follows:

(1) Power Unit

The power unit is a core component of a DSA and is primarily responsible for providing a relatively independent energy supply to each actuator. Considering the

diversity of power sources, energy-transmission modes, and actuator technologies in CM, three unified classification criteria are adopted: the type of primary energy source, the degree of decentralization and integration between the power unit and the actuator, and the terminal energy-transmission mode. Accordingly, five representative power-unit configurations are identified, as illustrated in Fig. 5, while their structural characteristics and applicability are summarized in Table 3. Representative application cases are discussed further in Section 3.

- **Electric-drive distributed hydraulic power-supply configuration:** A battery serves as the energy source, while electric motor–hydraulic pump units are distributed as independent power units to drive conventional hydraulic actuators through hydraulic circuits. This configuration retains the high load-bearing capability and actuation characteristics of conventional hydraulic systems while enabling distributed power supply and relatively independent energy delivery to individual actuators. It is therefore suitable for the electrification of conventional hydraulic CM and for decoupling multi-actuator systems. Its main limitation is that the power units and actuators remain physically separated; consequently, hydraulic pipelines and the working fluid still introduce transmission losses, resulting in a relatively limited degree of system integration [Niraula et al., 2018].
- **Electric-drive integrated EHA configuration:** A battery supplies electrical energy to a highly integrated, independent EHA unit comprising an electric motor, hydraulic pump, local hydraulic circuit, and actuator. Compared with the distributed hydraulic power-supply configuration, this architecture further shortens the hydraulic energy-transmission path while combining the high-power density of hydraulic actuation with independent control capability. It is particularly suitable for heavy-load and high-inertia operating conditions, as well as applications requiring high power density and fast dynamic response. Its principal engineering challenges include thermal management, sealing, additional mass, and the design complexity associated with a high degree of integration [Tobias et al., 2021].
- **Electric-drive direct EMA configuration:** A battery serves as the energy source, while an electric motor drives the load through a mechanical transmission mechanism, thereby eliminating the hydraulic power-transmission stage. This configuration features a short energy-conversion chain, high control accuracy, and no risk of hydraulic leakage, making it an important technological pathway toward P-ECM. At present, it is better suited to applications with relatively low power demand and limited impact loading, particularly where high efficiency, precise motion control, and clean operation are required. Its major engineering challenges include limited adaptability to heavy and impact loads, thermal management, reliability, and cost [Volvo Construction Equipment, 2017; Moog Inc., 2022; Bobcat Company, 2022].
- **ICE-based accumulator-assisted configuration (partially distributed):** An ICE–hydraulic pump unit serves as the primary power source, while accumulators provide local energy storage, energy recovery, and auxiliary power supply. This configuration can improve the utilization of regenerated energy without fundamentally altering the conventional hydraulic system, while retaining the high load-bearing capability of hydraulic machinery. It is therefore suitable for heavy-duty CM that continues to rely primarily on ICE-driven hydraulic systems. However, the main power source remains centralized, and actuator coupling and some throttling losses cannot be fully eliminated. Therefore, this configuration represents only a partial implementation of the distributed concept [Ge et al., 2017].
- **ICE-based distributed hydraulic power-supply configuration:** The conventional centralized ICE-based power source is reconfigured into multiple relatively independent ICE–hydraulic pump units, each supplying a dedicated actuator or functional module. This configuration enables power output to be matched to local load demand, reduces power mismatch associated with centralized supply, and improves actuator decoupling and dynamic response. It is suitable for applications that retain ICE-based power while requiring independent energy supply to multiple actuators. Its main limitations are the

increased complexity of system integration, coordinated control, and maintenance associated with multiple power units, together with the continued issues of fuel consumption and emissions [Wang et al., 2023a].

Table 3 Comparison of characteristics and applicability of five power-unit configurations.

Power-unit configuration	Key feature	Energy path	Advantages	Applications	Limitations
Electric-drive distributed hydraulic power supply	Distributed motor–pump units physically separated from hydraulic actuators	Battery → Motor → Pump → Hydraulic circuit → Actuator	Hydraulic compatibility; independent supply; high load capacity	Electrification ; multi-actuator decoupling	Hydraulic losses; low integration
Electric-drive integrated EHA	Motor, pump, local hydraulic circuit, and actuator highly integrated	Battery → Motor → Pump → Local circuit → Actuator	High power density; short path; independent control	Heavy-load, high-inertia applications	Thermal/sealing issues; integration complexity; added mass
Electric-drive direct EMA	Mechanical transmission without hydraulic power transmission	Battery → Motor → Mechanical transmission → Actuator	Short energy chain; high efficiency/precision; no leakage	Low-/medium-power, high-precision applications	Limited impact capability; thermal/reliability/cost issues
ICE-based accumulator-assisted configuration (partially distributed)	Central ICE–pump with local accumulator support	Fuel → ICE → Pump → Hydraulic system/Accumulator → Actuator	Mature hydraulics; energy recovery; heavy-load capability	Heavy-duty retrofit; hybrid systems	Partial distribution; actuator coupling; hydraulic losses
ICE-based distributed hydraulic power supply	Multiple independent ICE–pump units serving different actuators	Fuel → ICE units → Pumps → Actuators	Local power matching; reduced coupling; faster response	Multi-actuator ICE-powered machinery	More units; complex control/maintenance; emissions

(2) Actuator Unit

The actuator units can be categorized into linear drive and rotary drive types, as indicated by the purple region in Fig. 5.

- **Linear Drive:** Hydraulic cylinders are the most common linear actuators, widely used in heavy-load operations such as excavation and lifting due to their high load-bearing capacity. EMA, driven by electric motors through transmission mechanisms, achieve high-precision linear motion and are suitable for applications requiring greater motion accuracy [Bobcat Company, 2022].
- **Rotary Drive:** Hydraulic motors, as key rotary actuators within hydraulic systems, can deliver high torque output and are ideal for applications with high power demand.

(3) Control and Coordination Unit

The control and coordination unit constitutes the information-control layer of the DSA. Its core function is to enable coordinated operation among multiple relatively independent functional nodes through local sensing, node-level control and decision-making, network communication, and system-level coordination [Ding et al., 2018; Chen et al., 2010]. As illustrated in Fig. 5, each local control node performs fast closed-loop control and local optimization based on locally sensed states such as pressure, displacement, and velocity, while exchanging state, task, and control information with neighboring nodes or supervisory coordination nodes through communication networks such as Controller Area Network (CAN) and EtherCAT.

According to the allocation of decision-making authority and the mode of inter-node information exchange, the information-control network of a DSA can be generally classified into three typical topologies: hierarchical supervisory, peer-to-peer distributed, and hybrid distributed, whose structural characteristics and applicability are compared in Table 4. In the hierarchical supervisory topology, a supervisory node is responsible for task allocation and system-level optimization, while local nodes perform real-time closed-loop control. In the peer-to-peer distributed topology, coordinated decision-making is achieved through neighborhood information exchange and consensus mechanisms, thereby reducing dependence on a single control node. The hybrid distributed topology combines local autonomy with supervisory coordination to balance real-time performance, scalability, and global optimization. The architecture shown in Fig. 5 represents a hybrid information-control architecture characterized by local autonomy and supervisory coordination.

Table 4 Comparison of typical information-layer distributed control network topologies in DSA.

Control topology	Decision-making structure	Information exchange	Main characteristics
Hierarchical supervisory topology	Supervisory coordination + local closed-loop control	Communication between local nodes and the supervisory node	Clear global task coordination, but relatively dependent on the supervisory node
Peer-to-peer distributed topology	Autonomous decision-making by individual nodes	Direct information exchange among neighboring nodes	Strong scalability and resilience to single-point failures, but higher requirements for consensus and real-time communication
Hybrid distributed topology	Local autonomy + necessary supervisory coordination	Multi-level interaction among local, neighboring, and supervisory nodes	Balances real-time control, local autonomy, and system-level optimization; suitable for complex multi-actuator systems

On this basis, the fault-diagnosis module can utilize multi-node state information to perform local fault identification, isolation, and reconfiguration, thereby providing state information for subsequent cooperative and fault-tolerant control [Guo et al., 2022; Janbi et al., 2023]. Detailed methods for multi-actuator coordination, consensus control, network-induced delays, and fault-tolerant control are discussed further in Section 5.

(4) Energy Management Unit

The energy-management unit coordinates energy sharing and regeneration across electrical, mechanical, and hydraulic domains.

- **Energy Sharing Mechanism:** In DSA, energy sharing occurs across the electrical, mechanical, and hydraulic stages of the energy path, as illustrated in Fig. 5. Electrical energy sharing allocates battery power among electric motors through power scheduling and battery management [Fassbender et al., 2021]. Mechanical energy sharing distributes motor power through clutches or gearboxes according to time-varying demand. Hydraulic energy sharing allocates flow among actuator branches through devices such as flow dividers and priority valves [Wang et al., 2023b].
- **Energy Regeneration Mechanism:** Energy regeneration in CM mainly targets braking energy and boom potential energy, which together account for approximately 51% of the total recoverable energy [He et al., 2020]. Depending on the conversion path, recovered energy can be stored electrically in batteries or supercapacitors [Hao et al., 2023; Li et al., 2021; Hao et al., 2022], hydraulically in accumulators [Xia et al., 2018; Yang et al., 2023], or mechanically in flywheels [Cronk et al., 2021; Li et al., 2021]. These routes differ in conversion losses, controllability, storage density, and implementation complexity [Lin et al., 2017].

2.2 Characteristics of DSA

By decomposing traditional CSA into multiple independent modules, DSA achieve functional decoupling between power sources and actuators. This architecture

demonstrates significant advantages in terms of flexibility, SEE, fault tolerance, and response speed, as outlined below:

- **Decoupling of Power Sources and Actuators**

DSA adopt a decoupled design between actuators and power sources, enabling each actuator to independently obtain the required energy according to actual demand [Casoli et al., 2020]. This design effectively avoids the energy waste caused by throttling control and load disparities in traditional CSA, while significantly enhancing system flexibility. Under complex and highly variable operating conditions, actuators can be driven on demand, thereby markedly improving operational SEE and response speed.

- **Reduction of Throttling and Pressure Losses**

The decentralized design of DSA places power sources closer to the actuators, enabling short-path power supply that effectively reduces pipeline pressure losses and throttling losses, thereby significantly improving overall SEE [Wang et al., 2022].

- **Enhanced Fault Tolerance and Isolation**

Modularity and decentralization are core characteristics of DSA. When a failure occurs in a specific module, the remaining parallel modules can continue to operate independently, thereby reducing the impact of local faults on overall system performance and significantly enhancing fault tolerance and operational reliability [Zhu et al., 2024; Minav et al., 2018].

- **Flexible Energy Management Mechanisms**

As detailed in [Section 2.1.2](#), DSA improve SEE by coordinating energy allocation among electrical, mechanical, and hydraulic domains and by recovering otherwise dissipated potential and kinetic energy [Zhang et al., 2024; Quan et al., 2021].

- **Improved System Response Capabilities**

Independent power supply paths enable DSA to flexibly adjust power according to the real-time load of each actuator, thereby avoiding mutual interference among actuators [Guo et al., 2022]. Under high-load or impact operating conditions, this design significantly enhances system response speed and operational stability, ensuring efficient performance in complex working environments.

2.3 Classification of DSA

DSAs are not completely independent of traditional CSAs; rather, they have gradually evolved under the combined development of power-source electrification, energy-storage technologies, independent actuator drives, and modular design. To avoid classifying these architectures solely according to the chronological emergence of individual technologies, this review further distinguishes CM working-system architectures according to four principal features: the type and degree of centralization of the primary power source, the independent power-supply relationship between power units and actuators, the actuator-side control/actuation mode, and whether hydraulic energy-transmission media are retained. As illustrated in [Fig. 6](#), the six types represent typical architectural states along the evolution from centralized power supply toward actuator-level independent power supply and fully electric direct actuation. Their principal discriminative features and boundaries are summarized in [Table 5](#).

Type 1 represents the conventional centralized hydraulic architecture. A single ICE serves as the centralized prime mover, driving a hydraulic pump to generate a common hydraulic power supply, while a multiway valve meters and distributes flow to different actuators. In this architecture, all actuators share the centralized power source and hydraulic supply line, and no independent power unit is provided at the actuator side. Type 1 therefore serves as the baseline CSA from which the subsequent DSAs evolve.

Type 2 represents distributed volumetric control at the actuator side with a centralized prime mover. Mechanical power is still supplied centrally by a single ICE, whereas multiple hydraulic pumps or independent pump-controlled circuits are used to drive

different actuators, and local accumulators may be incorporated for energy recovery and auxiliary power support. Compared with Type 1, the key change is the transition from common valve-controlled flow distribution to relatively independent pump or volumetric control at the actuator side. However, the primary power source remains centralized, and no independent active auxiliary power branch is introduced. The “centralized power with distributed volumetric control system” discussed later in this review is representative of this category.

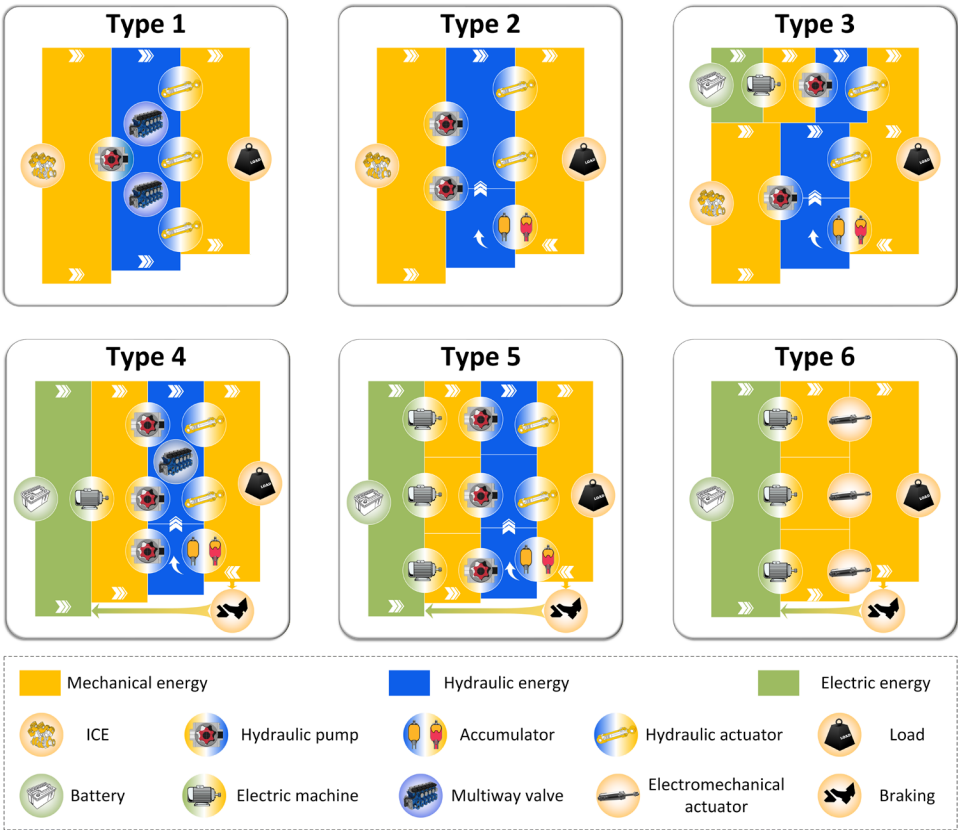


Fig. 6 Evolution and classification of six representative CM working-system architectures based on dominant energy-flow paths and actuator-level power-supply independence. Types 1–6 correspond respectively to centralized hydraulic power supply, centralized power with distributed volumetric control, hybrid centralized–distributed power supply, centralized electric–hydraulic power supply, actuator-level distributed electro-hydraulic drive, and fully electric distributed direct actuation. The six types represent a technological spectrum from centralized to increasingly distributed architectures rather than a strictly chronological or mutually exclusive classification.

Type 3 represents a hybrid centralized–distributed power-supply architecture. While retaining the centralized primary power source, additional local power branches, such as battery–motor units, accumulators, or other sources capable of independently contributing to load actuation, are introduced so that selected actuators can obtain active power from sources other than the central prime mover. Therefore, the fundamental boundary between Type 2 and Type 3 is not simply the presence of an energy-storage component, but whether an active distributed power branch capable of independently contributing to actuator drive is established. Such architectures generally combine centralized main power supply with local distributed actuation, as exemplified by hydraulic–electric dual-power systems and hybrid systems employing electrically active and hydraulically passive actuation.

Table 5 Discriminative features and boundaries of the six CM working-system architectures.

Type	Primary power source	Power-supply topology	Actuator-side control/actuation mode
Type 1	ICE	Centralized	Multiway-valve throttling control
Type 2	ICE	Centralized prime mover + multiple pumps/local storage	Distributed pump/volumetric control
Type 3	ICE + local electric-drive/energy-storage source	Hybrid centralized–distributed	Pump-controlled, electro-hydraulic, or electromechanical hybrid actuation
Type 4	Battery + single/centralized electric motor	Centralized electric–hydraulic supply	Centralized hydraulic actuation
Type 5	Battery + multiple motors/electro-hydraulic power units	Distributed	Independent pump-controlled/EHA-based electro-hydraulic actuation
Type 6	Battery + multiple electric motors	Distributed	EMA-based direct electromechanical actuation

Type 4 represents a centralized electric–hydraulic architecture. With the maturation of electric-drive technologies, the ICE is replaced by a battery–motor system, and the primary energy source changes from fuel to electricity. However, most actuators are still supplied by a centralized hydraulic system driven by a single electric motor and hydraulic pump. Therefore, the essential change in Type 4 is electrification of the prime mover rather than full decentralization of the power-supply topology. This configuration provides substantially improved prime-mover efficiency while establishing a technological basis for subsequent multi-motor and multi-power-unit distributed architectures.

Type 5 represents actuator-level distributed electro-hydraulic drive architecture. Taking advantage of the inherent multi-terminal power-supply capability of electric drives, the conventional centralized power source is further decomposed into multiple relatively independent motor–pump units, EHAs, or other electro-hydraulic drive units serving individual actuators or functional modules. Compared with Type 4, the principal discriminative feature is the transition from “one motor supplying multiple actuators” to “multiple power units independently supplying different actuators.” This enables further physical decoupling between power sources and actuators. Hydraulic media are still retained, but the energy-transmission paths are significantly shortened, while modularity and independent controllability are enhanced. Distributed EHA systems, independent pump-controlled systems, and hybrid centralized–distributed load-sensing (LS) systems reported in the literature all reflect this evolutionary direction.

Type 6 represents a fully electric distributed direct-actuation architecture. Building on Type 5, the hydraulic energy-conversion and transmission stages are completely eliminated. Batteries supply multiple electric motors, while EMAs directly drive the loads. The defining boundary of Type 6 is therefore the complete elimination of hydraulic media and the hydraulic power-transmission chain, resulting in a fully electric DSA in which electrical and mechanical energy dominate the entire energy path from the source to the actuator. This configuration represents the most extreme form of DSA development toward shortened energy-conversion chains and actuator-level independence.

It should be emphasized that the six architectures above are not strictly mutually exclusive categories, but rather representative architectural states and a technological spectrum describing the evolution of CM working systems from centralized toward increasingly distributed configurations. Types 2 and 3 exhibit clear hybrid centralized–distributed characteristics; Type 4 achieves electrification of the prime mover while retaining a predominantly centralized power-supply topology; and Types 5 and 6 further increase actuator-level independence in power supply and actuation. In practical systems,

centralized main power paths and local distributed actuator units may coexist. For example, EHAs or EMAs can be arranged in parallel with conventional LS systems to balance compatibility with existing architectures, power-capacity requirements, installation-space constraints, and energy-saving objectives.

Accordingly, the classification proposed in this review does not require every subsystem to strictly satisfy all characteristics of a single type. Instead, the principal architectural category is determined according to the dominant energy-flow path and the functional role of distributed power units within the system. With continued advances in electric drives, energy storage, intelligent control, and modular technologies, further hybrid and transitional configurations may emerge between adjacent categories. The subsequent sections combine representative studies and engineering cases to further discuss the application characteristics and evolutionary relationships of these architectures.

3 Research Progress on the DSA in CM

As one of the key pathways toward the electrification of hydraulic drives, the distributed concept in CM emerged after 2010. Researchers have explored various forms of distributed extensions based on traditional CSA, with excavators serving as the most representative study platform, gradually evolving into system configurations centered on EHA as the primary drive units. With continued advancements in electrification technologies, EMA have recently emerged as another promising distributed solution. However, due to high development costs [Fassbender *et al.*, 2021], EMA applications have largely been confined to prototype development within leading industrial enterprises. Representative system cases and the associated technological evolution are illustrated in Fig. 7.

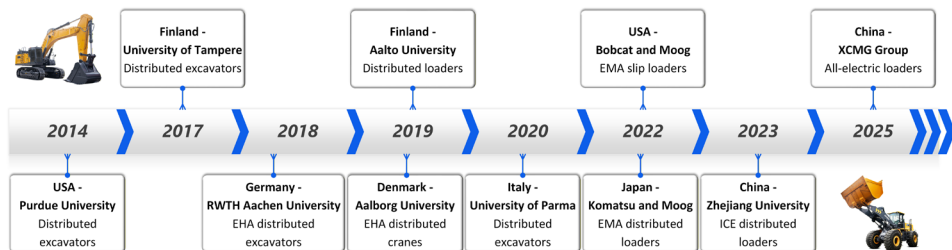


Fig. 7 Development of DSA in CM, showing representative system architectures and their evolution from centralized hydraulic power supply toward distributed electro-hydraulic and fully electric actuation.

The hydraulic power system of CM consists of a power end and an actuator end, with energy transmitted through hydraulic or electrical circuits. Accordingly, DSA can be structurally divided into power-end distribution and actuator-end distribution. Widely applied systems currently rely on a single power source to supply energy and use control valves to throttle and regulate multiple actuators. Since each actuator performs distinct operational functions, these systems already exhibit distributed throttling control capabilities at the actuator end and can be defined as “centralized power source with distributed throttling control systems,” corresponding to Type 1 in Fig. 6. Researchers have conducted multi-faceted studies on system architecture optimization, focusing on centralized and distributed expansions at both the power and actuator ends, as well as the evolution of different control strategies.

3.1 Centralized Power with Distributed Volumetric Control System

As is well known, throttling losses in control valves are one of the primary factors contributing to the low efficiency of throttling-controlled systems. Therefore, eliminating the valve assembly and adopting direct pump-flow control is considered an important alternative for achieving efficient hydraulic control. In 2014, Professor Monika at Purdue University proposed a multi-actuator hydraulic system based on flow control and pump

switching [Busquets et al., 2014; Busquets et al., 2015a], as shown in Fig. 8. This system was driven by a single ICE, which powered multiple hydraulic pumps via a belt linkage, supplying the cylinders and motors, respectively. Flow was directed to the target actuators through switching valves and distribution manifolds. This design not only enabled the combination of multiple pump flows to meet high-flow requirements but also effectively reduced the number of pumps needed for multi-actuator systems. Simultaneously, the research team developed an adaptive robust control strategy based on actuator prioritization [Busquets et al., 2015b], which was validated on both test benches and excavator platforms. The results demonstrated that this system architecture and control strategy exhibited reliable performance across multiple load conditions. This solution represents a typical “centralized power with distributed volumetric control system,” marking the first systematic application of the distributed concept in hydraulic systems for CM.

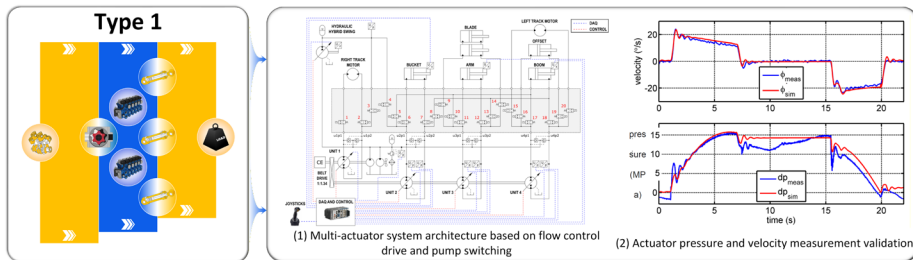


Fig. 8 Excavator architecture with centralized power supply and distributed volumetric control (Cited from [Busquets et al., 2015b]).

3.2 Centralized Power, Distributed Power and Distributed (Volumetric and electro-mechanical-hydraulic) Control System

As the power source in CSA, ICE is inherently limited in providing multi-port power output. Nevertheless, owing to their technological maturity and widespread application, extensive efforts have been made to explore distributed concepts based on centralized power sources with the aim of improving overall SEE. To address the low operating efficiency and significant energy losses of hydraulic excavators, Ref. [Ge et al., 2018] proposed a distributed variable-speed volumetric-drive hydraulic excavator architecture based on a centralized power source, in which accumulators were installed at the outlets of each pump for gravitational potential energy recovery and reutilization. This configuration achieved power decoupling during compound excavator operations, enabling each actuator to be independently regulated via volumetric control, thereby eliminating throttling losses induced by load mismatches. The system architecture is illustrated in Fig. 9. In addition, a coordinated control strategy combining segmented speed regulation and continuous displacement control was developed. Through the coordinated adjustment of rotational speed and displacement, pressure fluctuations under high-speed lifting conditions were effectively suppressed [Ge et al., 2017]. Experimental results demonstrated that the proposed architecture and control strategy achieved a maximum energy utilization efficiency of up to 71%, with an improvement in power-source efficiency exceeding 20%. From an architectural perspective, this system is characterized by centralized power supply, accumulator-based potential energy recovery and power reutilization, and distributed volumetric control, and can therefore be classified as a “centralized power, distributed power and distributed volumetric control system.” Overall, the proposed architecture and control strategy significantly enhanced the comprehensive SEE of the excavator.

In addition to integrating accumulators as “distributed power sources” with distributed volumetric control architectures for SEE improvement, accumulators have also been combined with EMA and hydraulic cylinders. By fully leveraging the high SEE of electromechanical transmission and the high-power density of hydraulic transmission, this combination forms a “centralized power, distributed power and distributed

electromechanical–hydraulic system” architecture, which further enhances the SEE of electro-hydraulic systems.

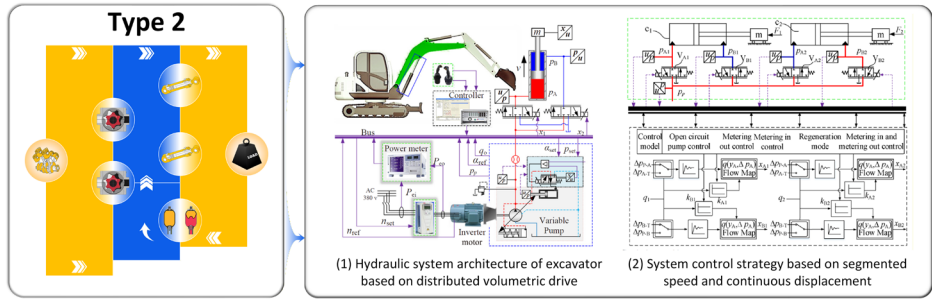


Fig. 9 Excavator architecture with centralized primary power, distributed auxiliary power, and distributed volumetric control (Cited from [Ge et al., 2017, 2018]).

① Hydraulic-electric dual-power-driven excavator boom system

To address throttling losses and gravitational potential energy dissipation during excavator operation, Ref. [Hao et al., 2023] proposed a novel excavator boom system architecture based on hydraulic-electric dual-power drive, as illustrated in Fig. 10. On the basis of this architecture, a control strategy centered on load state identification was developed. By employing an EMA for boom lifting and lowering control, energy losses associated with conventional throttling regulation were effectively avoided. In addition, the accumulator was directly connected to the hydraulic cylinder, which not only compensated for pressure losses in the hydraulic pump but also enabled gravitational potential energy recovery and reduced throttling losses caused by load differences among multiple actuators. Experimental results demonstrated that the proposed system reduced energy consumption by 66.1% and 38.1% under boom lifting/lowering and land leveling operating conditions, respectively, thereby significantly improving the overall SEE.

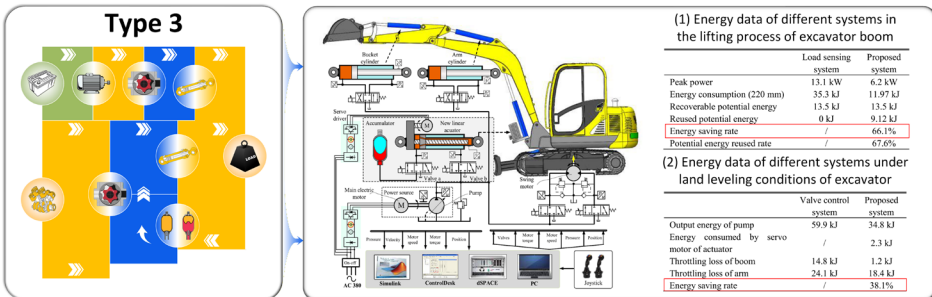


Fig. 10 Excavator system architecture (1) with centralized power, distributed power and distributed electro-mechanical-hydraulic control (Cited from [Hao et al., 2023]).

② Electro-hydraulic hybrid system for excavators based on electrically active drive and hydraulically passive drive

To achieve high-efficiency and high-performance control of the excavator boom lifting system, Fig. 11 presents a novel electro-hydraulic hybrid drive system composed of an electrically active drive and a hydraulically passive drive. In this configuration, the EMA was responsible for boom lifting and lowering control, while the high-power density combination of a hydraulic cylinder and an accumulator was employed to recover gravitational potential energy. This arrangement effectively mitigated the influence of the large-inertia boom during acceleration and deceleration, as well as the impact of rapid accumulator charging and discharging on the EMA. Experimental tests conducted on an excavator, with energy-consumption comparisons against a conventional LS drive system and a three-chamber hydraulic cylinder drive system, demonstrated that the proposed electro-hydraulic hybrid system reduced energy consumption by approximately 67% compared with the LS system [Li et al., 2021]. Compared with the three-chamber

hydraulic cylinder drive system incorporating potential-energy regeneration, the energy consumption was reduced by 31%, and the peak power was decreased by 15% [Qiao *et al.*, 2022]. In addition, the proposed system significantly reduced velocity overshoot and settling time, thereby enabling low energy-consumption and high-performance actuation of the excavator boom.

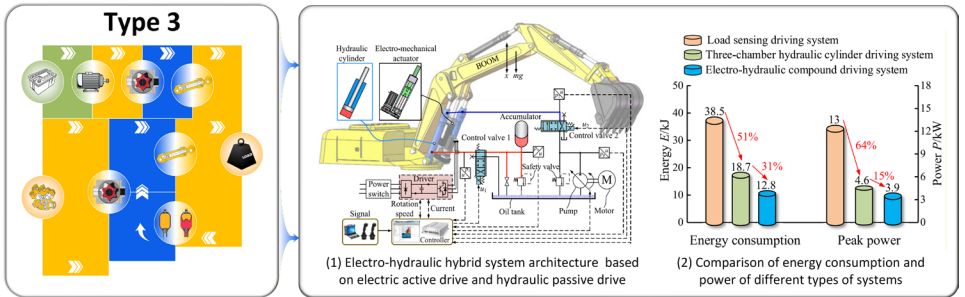


Fig. 11 Excavator system architecture (2) with centralized power, distributed power and distributed electro-mechanical-hydraulic control (Cited from [Qiao *et al.*, 2022]).

The continued maturation of electric-drive and energy-recovery technologies has supported the replacement of ICE prime movers by electric drives. Since 2020, major manufacturers have successively launched P-ECM products, as illustrated in Fig. 12. Taking electric wheel loaders as a representative example, electric drives can increase power-source efficiency from approximately 15% for conventional ICEs to about 90% [Lin *et al.*, 2020]. This substantial advantage drove rapid market growth, with sales in 2023 increasing by 209.9% year-on-year compared with 2022, representing substantial progress in both technological development and commercialization [China Construction Machinery Association, 2024b]. Such electric-drive-centered system configurations correspond to Type 4 in Fig. 6, in which the energy conversion path involves the motor converting electrical energy into mechanical energy to drive a hydraulic pump, generating hydraulic energy that is then delivered to actuators to produce linear or rotary motion for task execution. In addition, the integration of accumulator-based energy storage and regenerative braking further enhances overall SEE. At present, P-ECM still predominantly adopts a CSA in which a single electric motor drives multiple hydraulic actuators at the work end. However, the inherent multi-port power supply capability of EDT provides a critical foundation for the future realization of DSA in CM.



Fig. 12 Representative mass-produced P-ECM adopting the Type 4 centralized electric-hydraulic architecture (images obtained from official manufacturer websites).

3.3 Distributed Power and Distributed Volumetric Control System

With the continuous advancement of EDT, battery-powered P-ECM has gradually matured. The use of batteries as power sources addresses the limitation of ICE in providing multi-port energy supply, making it feasible to drive hydraulic systems through a “distributed power” approach. Consequently, several universities have explored CM system architectures based on the “distributed power and distributed volumetric control system” configuration.

The research team at RWTH Aachen University developed a distributed hydraulic system that combined high-speed internal-gear pumps with preloaded pressure, achieving pump speeds of up to 10,000 rpm [Roth et al., 2019]. Based on this system, a “distributed power and distributed volumetric control” excavator configuration was constructed, as shown in Fig. 13. This design aimed to reduce throttling and line losses inherent in conventional CSA, thereby enhancing overall SEE. The team also compared the differences between direct electric drive and mechatronic high-speed drive in terms of power density and cost-effectiveness, proposed design concepts for both open- and closed-circuit hydraulic systems, and validated the approach using an EHA prototype [Roth et al., 2019]. Another study employed a structured simulation method to compare the performance of different EHA circuits, showing that a circuit with a four-solenoid valve topology performed better in preventing cylinder retraction oscillations [Figge et al., 2023]. Furthermore, the research highlighted that increasing pump speed is crucial for enhancing system power density: when the internal-gear pump speed increased from 4,000 rpm to 9,000 rpm, power density improved significantly, and appropriate adjustment of boost pressure effectively mitigated cavitation issues [Pietrzyk et al., 2018]. Overall, these studies indicate that, through optimization of system design and manufacturing processes, the “distributed power and distributed volumetric control” system exhibits substantial potential for improving compactness, SEE, and power density.

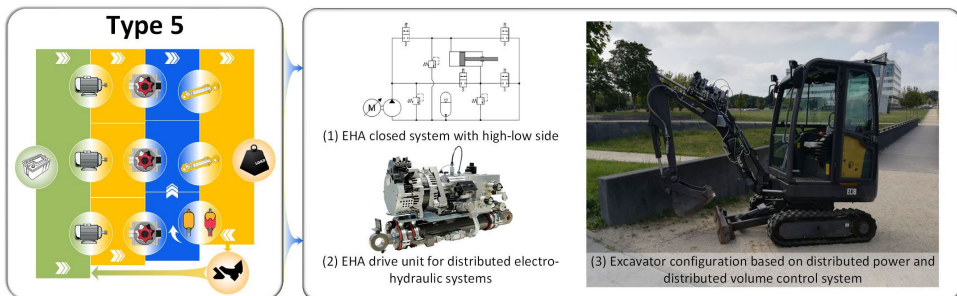


Fig. 13 Excavator system architecture (1) with distributed power and distributed volumetric control system (Cited from [Tobias et al., 2021; Roth et al., 2019]).

Around the EHA-driven system, Minav and colleagues systematically explored its application potential within DSA. EHA enable high-precision displacement control without the need for conventional valve-based regulation. Danaee et al. demonstrated their effectiveness in mini excavators [Danaee et al., 2018], with the system configuration illustrated in Fig. 14. Although EHA offer significant SEE advantages, they also introduce additional mass and structural stress. Finite element analysis indicated that despite an approximate 15% increase in system mass, structural integrity remained largely uncompromised, while overall energy utilization improved by roughly 12% [Niraula et al., 2018]. Furthermore, simulations conducted by Zhang et al. showed that, compared to traditional LS systems, a DSA employing EHA could achieve a total efficiency of up to 71.3% under typical excavator operating cycles [Zhang et al., 2017a; Zhang et al., 2017b]. Subsequent studies proposed a single-dual pump hybrid system incorporating an active disturbance rejection controller, which substantially enhanced system response speed and trajectory tracking accuracy while effectively reducing energy consumption [Zhang et al., 2017a; Zhang et al., 2023]. Collectively, these findings

indicate that the “Distributed Power and Distributed Volumetric Control System” demonstrates substantial potential for SEE improvements in P-ECM, though further optimization is needed in terms of system weight reduction and control precision.



Fig. 14 Excavator system architecture (2) with distributed power and distributed volumetric control system (Cited from [Zhang et al., 2017a]).

Casoli and colleagues systematically investigated the application of distributed power combined with distributed volumetric control systems in 9-ton excavators. Building on Minav’s research, they refined the mathematical model of conventional LS systems to conduct an in-depth analysis of the SEE performance of EHA-based systems. The study consisted of two parts: the first evaluated the feasibility of replacing traditional valve-controlled hydraulic systems with EHA, with the system configuration illustrated in Fig. 15. The results indicated that, by eliminating directional valves and pressure compensators, EHA could significantly reduce hydraulic energy losses while optimizing mechanical power demand and ICE displacement [Casoli et al., 2020]. In certain configurations, the system could achieve higher efficiency and a more compact design by reducing engine displacement, simultaneously decreasing engine mass and size, thereby partially offsetting the additional impact of the introduced electrical components. The second part further examined the performance differences between open-loop and closed-loop EHA arrangements. Findings demonstrated that closed-loop EHA systems offered superior energy-saving performance compared to open-loop configurations and, due to fewer components, also excelled in system compactness, weight, and cost [Casoli et al., 2022]. This research provides crucial insights for enhancing SEE in CM and offers significant value for optimizing overall system performance while reducing fuel consumption.

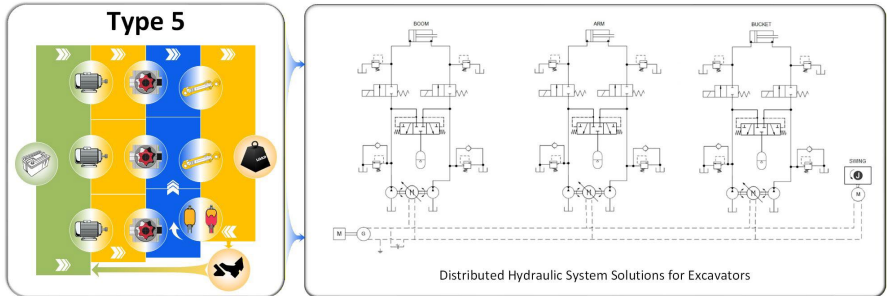


Fig. 15 Excavator system architecture (3) with distributed power and distributed volumetric control system (Cited from [Casoli et al., 2020]).

To address the significant energy losses inherent in traditional hydraulic systems, Yuan et al. proposed a distributed independent EHA control system, as shown in Fig. 16. The system employed a closed-loop pump-controlled configuration and introduced a speed control strategy based on mode switching and pump mapping to address the inlet–outlet flow mismatch caused by asymmetric cylinders in closed-loop operations, thereby effectively enhancing system performance [Yuan et al., 2024]. Furthermore, by analysing the impact of signal delay on system response characteristics, the signal transmission of

the switching valve control circuit was optimized, resulting in a 16.6% improvement in system response speed under a 0.1 s signal delay. To minimize installation power requirements and spatial footprint, a shared accumulator scheme between the boom and arm circuits was proposed, with optimal system performance observed at a precharge pressure of 20 bar. Experimental results indicated that the SEE was significantly increased from 19.19% to 59.56%, providing a novel technical pathway for energy saving and performance optimization in P-ECM.

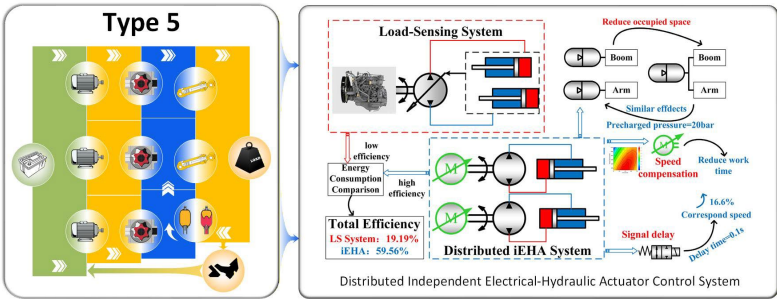


Fig. 16 Excavator system architecture (4) with distributed power and distributed volumetric control system (Cited from [Yuan et al., 2024]).

To address the low SEE caused by load differences while minimizing modifications to the existing system layout, a hybrid LS system integrating distributed and centralized concepts was proposed [Mu et al., 2025], as shown in Fig. 17. The system balanced load disparities among multiple actuators and enabled efficient potential energy recovery by arranging distributed actuators (EHA or EMA) in parallel with conventional hydraulic cylinders. Considering the four-quadrant load characteristics of CM, a motor torque controller was designed for load balancing and energy recovery of the distributed actuators. In addition, parameter design guidelines for the distributed actuators and control valves were developed to achieve an optimal trade-off between energy-saving capability and constraints on power capacity or installation space. Experimental results demonstrated that, when EHA was employed as the distributed actuator, battery energy consumption was reduced by 11.4%; when EMA was used, battery energy consumption was reduced by 13.6%, while the motion performance remained comparable to that of a conventional LS system. Moreover, the proposed system significantly reduced energy losses in the pressure compensator and effectively recovered potential energy, exhibiting substantial energy-saving potential.

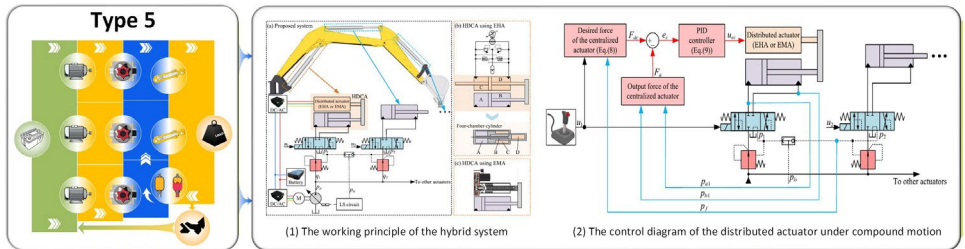


Fig. 17 Excavator system architecture (5) with distributed power and distributed volumetric control system, in which distributed EHA/EMA units operate in parallel with conventional hydraulic actuators to enable load balancing and potential-energy recovery (Cited from [Mu et al., 2025]).

By integrating the hydraulic power unit and the hydraulic cylinder into a compact module, EHA offers advantages such as high SEE and structural compactness; however, compared with conventional valve-controlled hydraulic systems, the additional mass introduced by EHA may reduce the effective payload capacity of cranes. Ketelsen investigated the impact of distributed EHA systems on the payload capacity of articulated boom cranes [Ketelsen et al., 2019]. By analysing the influence of different hydraulic

circuit configurations on the mass of independent hydraulic cylinders, the study found that the mass of an EHA unit was approximately 37% to 50% higher than that of a conventional hydraulic cylinder, which could lead to a reduction in effective payload capacity of about 0.8-3%. Therefore, the acceptability of distributed EHA applications in crane operations depends on the machine’s primary operating range.

Compared with excavators, wheel loaders employ wheeled traveling systems, resulting in a relatively lower proportion of hydraulic power in the overall energy consumption. Intuitively, this characteristic appears to limit the potential for achieving substantial research breakthroughs. Nevertheless, Ref. [Turunen et al., 2018] conducted an experimental study on a volume-controlled DSA applied to a mining wheel loader, in which the boom and bucket systems were systematically tested and comparatively analysed against a conventional LS system. The results demonstrated that the proposed DSA exhibited significantly higher SEE than the LS system across a wide range of operating conditions. In particular, under low-speed and light-load conditions, the efficiency improvement exceeded twentyfold, while under high-speed conditions the system also showed pronounced advantages. As one of the few studies focusing on DSA for wheel loaders, this work provides valuable insights and an important reference for the distributed design and SEE enhancement of loader hydraulic systems.

To address the SEE bottleneck arising from supplying multiple functions with a single hydraulic pump in CM, a representative solution is the introduction of power-sharing mechanisms with multiple energy pathways. Ref. [Wang et al., 2023b] proposed a power-sharing EHA system based on a planetary gear set, as illustrated in Fig. 18. By optimizing key parameters and incorporating a minimum power consumption strategy, the system enabled motor downsizing while improving overall SEE. Simulation results indicated that the proposed system could significantly reduce the total installed motor power requirement, achieving a 37.0% reduction compared with the original system. In addition, considering the centralized characteristics of ICE, a DSA based on a hydraulic free-piston engine was proposed. By tailoring power output to each function and adopting a digital LS control approach, the system achieved independent control and responsive operation of individual functional modules while maintaining high SEE [Wang et al., 2023a; Wang et al., 2022]. Experimental results demonstrated that the system could sustain an average pressure margin of 20 bar under inherent pressure fluctuations and realize continuous load flow.

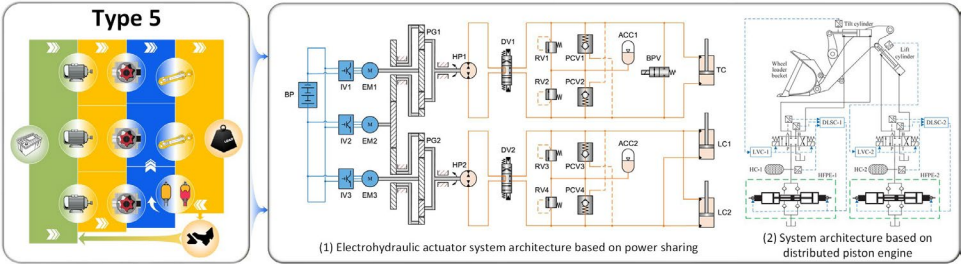


Fig. 18 Wheel loader system architecture with distributed power and distributed control system, illustrating power-sharing among multiple energy pathways and independent power regulation of functional modules (Cited from [Wang et al., 2023b; Wang et al., 2023a]).

Current electric wheel loaders lack energy-saving optimization strategies that explicitly address variations in traction demand during the bucket preparation phase, with particularly prominent issues related to tire slip and SEE. To this end, Ref. [Fei et al., 2023] introduced a “three-motor technology” and a “dual-motor distributed drive optimization strategy.” This approach employed three electric motors to jointly drive the loader, among which two motors independently drove the front and rear wheels, forming an industry-first series-connected dual-motor drive configuration, as shown in Fig. 19. By comparatively analysing three operating modes—front-motor drive, rear-motor drive, and dual-motor drive—the study found that when the bucket was not in contact with the

ground, the single-motor mode achieved the highest SEE, whereas when a positive normal force was generated between the bucket and the ground, the dual-motor drive mode exhibited superior performance.

In addition, for soil-engaging operations, the study employed a load estimation method combining singular value decomposition with unscented Kalman filtering, and optimized torque distribution using an improved particle swarm algorithm. By taking tire load rate as the optimization objective and imposing anti-slip boundary conditions, the approach effectively reduced wheel slip while significantly enhancing traction and SEE [Gao et al., 2022]. Experimental results fully validated the effectiveness of this distributed drive optimization strategy, providing an innovative technical solution for high-efficiency, energy-saving, and intelligent control in electric wheel loaders, particularly demonstrating superior performance under complex operating conditions.

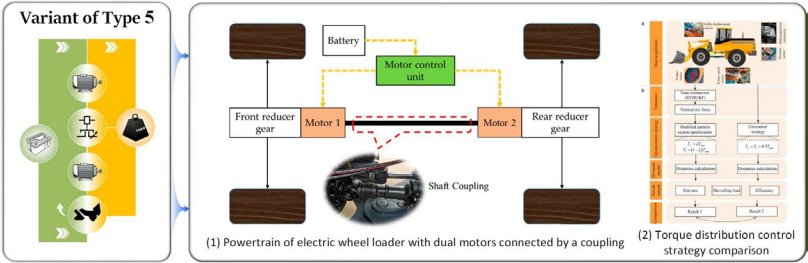


Fig. 19 Dual-motor DSA and control strategy for electric wheel loaders (Cited from [Fei et al., 2023; Gao et al., 2022]).

3.4 Distributed Power with Distributed Electromechanical Control System

With the rapid development of electrification and intelligent technologies in CM, the exploration of alternatives to traditional hydraulic drives has gained momentum. Solutions that employ batteries as the power source and fully replace hydraulic actuators with EMA enable the development of fully electric “distributed power with distributed electromechanical control systems,” representing the extreme application of DSA. Fig. 20 illustrates P-ECM equipped with EMA drive units, as developed by Volvo [Volvo Construction Equipment, 2017], Komatsu [Moog Inc., 2022], Bobcat [Bobcat Company, 2022], and XCMG [XCMG Group, 2025]. Compared to hydraulic drives, EMA convert electrical energy directly into mechanical energy, reducing intermediate energy conversion losses and thereby enhancing SEE. Moreover, EMA operate independently of hydraulic fluid, eliminating leakage risks and offering improved environmental performance. Leveraging electronic control, EMA provide high precision and rapid response capabilities, meeting the stringent dynamic performance requirements of CM. In addition, the flexible and compact layout of EMA facilitates space-saving designs and increases structural design freedom.



Fig. 20 P-ECM with a “distributed power and distributed electromechanical control system” configuration (Cited from [Volvo Construction Equipment; Moog Inc; Bobcat Company; XCMG Group, 2025])

4 Research Progress on Drive Units for DSA in CM

EHA and EMA represent two major types of independent drive units for DSA in CM. EHA convert electrical energy into mechanical output through an electric motor–hydraulic pump–hydraulic actuator energy-conversion chain. This architecture enables independent power supply at the actuator level while retaining the high-power density and heavy-load capability of hydraulic actuation. In contrast, EMA directly drive the load through an electric motor and mechanical transmission, thereby shortening the energy-conversion chain and offering high control accuracy, elimination of hydraulic leakage, and convenient integration into P-ECMs. Because the two drive technologies differ fundamentally in their energy-conversion paths, power density, mechanical compliance, and heat-generation characteristics, their relative performance strongly depends on load level, impact intensity, continuous operating duration, and control-accuracy requirements. Therefore, a generally applicable assessment cannot be derived solely from individual efficiency measurements or isolated experimental cases. This section first establishes a unified evaluation framework to clarify the application boundaries of EHA and EMA, and then reviews their respective research foundations and engineering application status.

4.1 Unified Evaluation and Application Boundaries of EHA and EMA

From the perspective of energy conversion, EMA directly drive the load through mechanical transmission mechanisms such as gear reducers, ball screws, or planetary roller screws. Owing to the reduced number of conversion stages, EMA can provide high transmission efficiency and control accuracy under suitable operating conditions [Tolomatic, n.d.]. EHA, by contrast, involve electrical–mechanical–hydraulic–mechanical energy conversion. Nevertheless, hydraulic transmission can generate large output forces within a relatively compact volume, while functions such as load-adaptive pressure regulation, energy regeneration, and load holding can improve system-level energy efficiency. Existing comparative studies indicate that, under certain heavy-duty design conditions, EHA may offer advantages over EMA with equivalent output capability in terms of system efficiency, mass, and cost, as illustrated in Fig. 21 [Rydberg *et al.*, 2015]. However, such conclusions are strongly dependent on actuator sizing, load spectrum, and duty cycle and should therefore not be generalized to all operating conditions. Other studies have also reported that increasing EMA output capability may result in larger actuator dimensions, greater thermal loading, and higher cost [Maurus *et al.*, 2017; Hagen *et al.*, 2020].

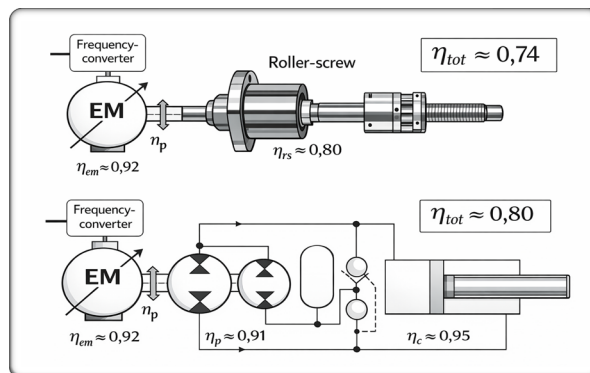


Fig. 21 Comparison of SEE between EHA and EMA under the representative actuator design conditions reported by Rydberg *et al.* (2015).

To avoid evaluating the two technological routes using a single performance indicator, this study compares EHA and EMA within a common framework covering system efficiency, power/force density, mass and packaging, cost, shock-load adaptability, control performance, thermal behaviour, lifecycle reliability, and engineering maturity, as summarized in Table 6.

Table 6 Comparison of Key Performance Characteristics of EHA and EMA.

Evaluation Metric	EHA	EMA
SEE	Relatively high; supports energy regeneration	High; short energy-conversion chain
Power/force density	High	Moderate; increasingly constrained at higher loads
Mass and packaging	More advantageous under heavy-load conditions; integrated auxiliary components add mass	Compact at low loads; size and mass increase under heavy loads
Cost	Mature technology; relatively low	Relatively high
Shock/overload adaptability	Strong	Relatively weak; additional buffering is generally required
Control accuracy	High	High
Thermal characteristics	Temperature-sensitive hydraulic fluid; distributed heat generation	Concentrated heat generation; constrained under continuous high-load operation
Lifecycle reliability	Relatively mature; affected by seal wear, leakage, and fluid degradation	Long-term durability under heavy-duty conditions remains insufficiently validated
Maintenance and environmental performance	Higher hydraulic maintenance requirements; leakage risk	No hydraulic leakage; cleaner operation
Engineering maturity	High; well-established basis for heavy-duty applications	Lower; currently dominated by prototypes and limited applications

It should be emphasized that no universally applicable fixed power threshold exists between EHA and EMA. The continuous and peak power capabilities of either actuator depend strongly on the transmission configuration, gear ratio, stroke, duty cycle, and cooling conditions. A representative actuator-selection study reported that, for the configurations and operating conditions considered, the efficiency and controllability advantages of EMA were primarily observed when continuous transmitted power remained below approximately 2 kW [Hagen et al., 2020]. This value, however, should be regarded as a result for a specific actuator design rather than a universal upper limit for EMA. With continued advances in high-torque-density electric motors, planetary roller screws, and power electronics, the applicable power range of EMA continues to expand.

Accordingly, defining the application boundary according to power demand, load characteristics, and duty cycle is more meaningful from an engineering perspective than adopting a single power threshold. Under current technological conditions, EHA are generally better suited to CM implements involving medium-to-high power demand, high output force, heavy loads, large inertia, severe impact loading, and frequent load variations. EMA are more suitable for applications characterized by low-to-medium continuous power, relatively mild impact loading, and high requirements for motion-control accuracy, cleanliness, leakage-free operation, and electrical integration [Tolomatic, n.d.; Maurus et al., 2017; Hagen et al., 2020; Huang et al., 2024; Qiao et al., 2024]. These boundaries are not fixed, but evolve continuously with advances in electric motors, mechanical transmissions, and impact-mitigation technologies.

From the perspective of current engineering maturity, EHA still possess a more extensive theoretical, prototype, and experimental foundation in heavy-duty mobile machinery, whereas EMA have not yet demonstrated a comprehensive capability to replace EHA in heavy-duty CM applications. A recent review of P-ECM also indicated that large-scale industrial deployment remains at an early stage and that the reliability of key actuation components remains an important issue for further engineering implementation [Huang et al., 2024]. Therefore, the following subsections separately examine the established research foundation of EHA and the current engineering status and key limitations of EMA.

4.2 Research Progress of EHA Drive Units in CM

In the engineering field, operating environments are typically complex and harsh, posing significant challenges to any type of drive unit. In recent years, researchers have extensively investigated the performance of EHA under extreme temperatures, heavy loads, high-precision requirements, and high-efficiency conditions in demanding CM applications, and have systematically evaluated the suitability of EHA drive units from multiple perspectives.

4.2.1 Extreme Temperatures and Thermal Behavior

CM commonly operates under low temperatures, high temperatures, and large ambient-temperature variations. Changes in fluid viscosity, pump leakage, and local heat generation can therefore significantly affect both the efficiency and dynamic behaviour of EHA. Building on previous studies of distributed pump-controlled excavators, Järf *et al.* introduced an accumulator to compensate for the flow mismatch between the two chambers of an asymmetric hydraulic cylinder, increasing SEE by approximately 30% [Järf *et al.*, 2016]. However, the high level of integration in EHA also results in stronger thermal coupling among the electric motor, hydraulic pump, valves, and working fluid [Zhai *et al.*, 2022].

Minav *et al.* and Karlén *et al.* developed thermo-hydraulic coupled models and used thermal imaging to investigate the characteristics of direct-driven hydraulic systems under different ambient temperatures [Minav *et al.*, 2019; Karlén *et al.*, 2016]. Their results demonstrated that temperature-induced variations in fluid viscosity can significantly affect pump leakage, friction losses, and overall system efficiency, indicating that appropriate fluid selection and thermal management are essential for maintaining stable EHA performance. Further studies have shown that, by accounting for the temperature dependence of pump volumetric efficiency and system parameters, flow compensation and robust control can be employed to mitigate temperature-induced degradation in actuator dynamic performance [He *et al.*, 2026]. Consequently, research on EHA for CM operating over wide temperature ranges is gradually evolving from isolated efficiency optimization toward integrated thermo-hydraulic-control design.

4.2.2 Adaptability to Heavy Loads, Large Inertia, and Impact Conditions

Heavy loads, large inertia, and strongly varying loads are important characteristics that distinguish CM working systems from conventional industrial servo systems. Altare *et al.* proposed compact EHA circuits with load-holding capability and improved system efficiency and power density through novel hydraulic-pump designs [Altare *et al.*, 2015; Altare *et al.*, 2014]. Subsequently, various EHA architectures incorporating energy regeneration and servo motor–pump units were developed, further improving load matching, dynamic response, and energy-efficient operation [Qu *et al.*, 2021; Zhao *et al.*, 2016]. Compared with rigid mechanical transmission systems, EHA can accommodate sudden external load variations through hydraulic pressure regulation, circuit protection, and a certain degree of hydraulic compliance. They therefore exhibit strong engineering competitiveness in applications that simultaneously require high output force, large-inertia actuation, and rapid response. This characteristic is also an important reason why EHA have been investigated relatively early for excavators, lifting equipment, and other heavy-duty machinery.

Nevertheless, the high-power density of EHA does not eliminate structural trade-offs. Integrating the pump, electric motor, and hydraulic components at the actuator level inevitably increases local actuator mass. For example, a study on knuckle-boom cranes reported that self-contained EHA units may be approximately 37%–50% heavier than conventional hydraulic cylinders, potentially reducing effective payload capacity by approximately 0.8%–3% [Ketelsen *et al.*, 2019]. Therefore, the engineering applicability of EHA must still be evaluated by jointly considering output force, installation location, mass distribution, and overall machine payload.

4.2.3 High Efficiency and Energy Regeneration

Reducing valve-throttling losses is one of the principal advantages of EHA over conventional centralized hydraulic systems. Padovani *et al.* proposed a self-contained single-rod EHA incorporating passive load-holding capability, a sealed reservoir, and energy-regeneration functions as shown in Fig. 22 [Padovani *et al.*, 2019]. Building on this architecture, Hagen *et al.* developed a DSA consisting of multiple EHAs and experimentally demonstrated it in offshore drilling equipment. Through regenerated-energy sharing and storage, energy consumption was reduced by as much as 83.44% compared with a conventional valve-controlled solution [Hagen *et al.*, 2019a; Hagen *et al.*, 2018; Hagen *et al.*, 2019b].

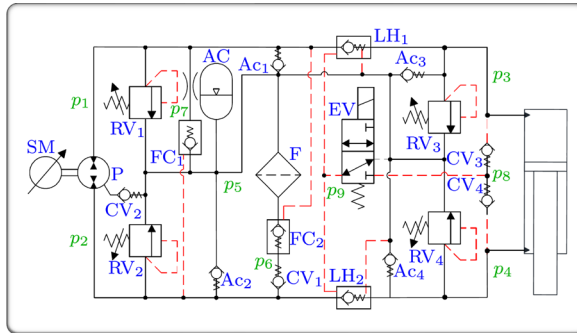


Fig. 22 DSA based on multiple self-contained EHA units, enabling independent actuator control and system-level energy regeneration and sharing (Cited from [Padovani *et al.*, 2019])

Comparative experiments with conventional LS systems have further shown that self-contained EHA can improve motion-control performance, including position tracking, while substantially reducing energy consumption through independent pump control and energy regeneration [Hagen *et al.*, 2019c; Hagen *et al.*, 2018]. The efficiency advantage of EHA therefore does not arise merely from improving the efficiency of individual hydraulic components. Instead, it results from system-level restructuring through elimination of centralized throttling, shortening of hydraulic energy-transmission paths, and actuator-level energy regeneration.

4.2.4 Advanced Control and System Integration

As EHA are increasingly applied to high-dynamic, wide-operating-range, and high-precision tasks, actuator performance has become increasingly dependent on system modelling, parameter identification, and advanced control algorithms. Moghaddam *et al.* applied fractional-order proportional–integral–derivative control with parameter optimization to an excavator EHA to improve dynamic performance [Moghaddam *et al.*, 2021]. Ouyang *et al.* developed a pressure-feedback-based model predictive control (MPC) strategy for a heavy-duty EHA and combined it with multi-objective optimization and state-observation methods to improve transient performance under heavy-load conditions [Ouyang *et al.*, 2023].

Overall, EHA research has progressively evolved from early investigations of actuator configuration and energy efficiency toward integrated optimization of structure, thermal behaviour, control, and energy management. Their high-power density, strong heavy-load adaptability, and mature hydraulic engineering foundation make EHA one of the more technologically mature actuator-level distributed-drive solutions currently available for CM [Helbig *et al.*, 2016; Jalayeri *et al.*, 2015; Ketelsen *et al.*, 2019]. Nevertheless, thermal management, sealing reliability, additional local mass, and the design and maintenance complexity associated with high integration remain important issues for further engineering development [Michel *et al.*, 2012; Schmidt *et al.*, 2017].

4.3 Engineering Status and Key Limitations of EMA Drive Units

EMA directly drive loads through electric motors and mechanical transmission mechanisms, thereby eliminating valve-throttling losses and hydraulic-fluid leakage. Their short energy-conversion path, high positioning accuracy, and convenient electrical integration make them an important potential actuation technology for P-ECMs. Existing studies have demonstrated their energy-saving potential in relatively low-impact applications such as lifting platforms. For example, Stawiński *et al.* investigated the replacement of conventional hydraulic actuation with EMA in a scissor lift through simulation and experimental testing and reported an approximately 35%–50% reduction in energy consumption over the investigated duty cycles [Stawiński *et al.*, 2023]. Meanwhile, manufacturers including Volvo, Komatsu/Moog, Bobcat, and XCMG have explored various fully electric actuator prototypes and products, as discussed in [Section 3](#).

However, these demonstrations do not necessarily indicate that EMA can already satisfy all requirements of typical heavy-duty CM working systems. Operations such as digging, collision, rapid braking, and reversal of large-inertia loads can generate substantial transient and peak loads. In EMA, components such as gear reducers, screws, and bearings form a relatively rigid mechanical load path, allowing transient impacts to be transmitted more directly to critical mechanical contact interfaces.

To address this problem, Qiao *et al.* proposed an impact-resistant EMA incorporating a disc-spring compliance mechanism and a hydraulic buffering mechanism. Experimental results showed that, under the same sudden-load condition, the peak impact force transmitted through the electromechanical drivetrain could be reduced by more than 50% [Qiao *et al.*, 2024]. On the one hand, this demonstrates that compliant and buffering mechanisms can improve the impact adaptability of EMA; on the other hand, it also indicates that impact resistance remains a key issue that must be specifically addressed when extending EMA toward heavy-duty CM applications.

As output force and continuous power increase, load-carrying capacity and service life of the mechanical transmission become increasingly critical. Planetary roller screws, which are commonly considered for high-load EMA, may experience progressive degradation in load-carrying capacity, transmission accuracy, and efficiency due to the coupled effects of temperature rise, vibration, rotational speed, and load [Chen *et al.*, 2024]. Under cyclic loading, non-uniform roller–screw contact loads, wear, and manufacturing errors can further affect operational stability and service life [Xing *et al.*, 2024]. It should be noted that these studies mainly concern reliability at the mechanical-transmission-component level and do not constitute full-machine validation under CM operating conditions. Nevertheless, they identify important lifecycle issues that EMA must address for long-term heavy-duty operation.

The current engineering status of EMA in CM can therefore be summarized as follows: their advantages in high efficiency, high control accuracy, and hydraulic-free operation have been demonstrated, and their application is gradually extending from light-duty and low-to-medium-power systems toward higher-load applications; however, long-term industrial validation in heavy-duty CM remains limited. Most publicly available studies still focus on proof-of-concept prototypes, individual actuators, or specific duty cycles, while systematic data remain scarce for long-term repeated impacts, dust contamination, wide temperature ranges, continuous high-load operation, and lifecycle reliability under multi-actuator operation. Existing reviews of P-ECM similarly indicate that large-scale engineering deployment remains at a relatively early stage [Huang *et al.*, 2024].

Potential approaches for extending EMA toward heavy-duty applications include increasing electric-motor torque density and mechanical-transmission load capacity, improving load sharing and lubrication in planetary roller screws, incorporating compliant or buffering mechanisms, strengthening thermal management under high heat-flux conditions, and using multi-source condition monitoring and predictive maintenance

to improve the lifecycle reliability of critical mechanical transmission components [Kang et al., 2024]. Although these approaches provide clear directions for improvement, whether they can simultaneously achieve an acceptable engineering balance among mass, cost, power density, impact reliability, and service life has not yet been sufficiently demonstrated.

In summary, at the current level of technological maturity, EHA retain higher overall applicability for CM working systems characterized by heavy loads, large inertia, and complex duty cycles, primarily because of the high-power density of hydraulic transmission, stronger adaptability to impact loading, and more mature engineering experience. EMA offer considerable potential in terms of energy efficiency, control accuracy, leakage-free operation, and fully electric integration; however, their widespread use as replacements for heavy-duty working actuators still depends on further advances in power density, impact mitigation, thermal management, mechanical-transmission durability, and long-term reliability validation. Therefore, at the present stage, EHA and EMA should not be regarded simply as mutually substitutable technologies, but rather as two distributed-drive routes with distinct operating-condition boundaries and different levels of engineering maturity.

5 Advances in Control Research for DSA in CM

Building on the information-control layer introduced in Section 2, the control paradigm of DSA shifts from single centralized control toward multi-node local control and system-level coordination. Unlike CSA, in which state processing and task allocation are handled by a single control core, local nodes in DSA must maintain fast local closed-loop control while achieving cross-node coordination through hierarchical supervision, neighborhood information exchange, or hybrid control topologies [Liang et al., 2024; Yong et al., 2023]. Consequently, the control performance of DSA depends not only on the tracking accuracy of individual actuators, but also on inter-node communication, synchronization and consensus, task allocation, and changes in network topology.

Current research mainly focuses on four aspects: (1) cooperative and synchronized control of multiple subsystems; (2) drive, braking, and energy-allocation control; (3) steering and motion control; and (4) fault diagnosis and fault-tolerant control. Together, these research directions reflect the evolution of DSA from local autonomous control toward multi-node cooperative optimization. The main research topics are illustrated in Fig. 23, and the corresponding advances are discussed below.

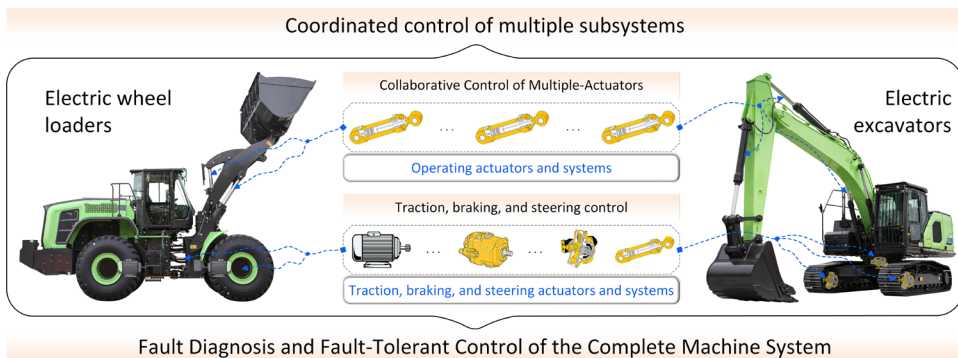


Fig. 23 Control issues in DSA for CM: Case studies of wheel loaders and excavators.

5.1 Cooperative Control of DSA

Distributed cooperative control has been widely applied to multi-agent systems and high-risk task scenarios, including robot swarms [Xu, P.J. et al., 2024], heavy multi-actuator machinery [Zhai et al., 2022; Wang et al., 2025], and unmanned aerial vehicle formations [Su et al., 2023]. Cooperative control among multiple subsystems in DSA can

be regarded as a further extension of multi-agent cooperative-control concepts within a single machine. Its core objective is to achieve synchronized operation among relatively independent actuation nodes through local sensing, neighbourhood information exchange, and cooperative decision-making.

Electromechanical-hydraulic systems in CM are affected by external load disturbances, cooperative-mode switching, actuator dynamic mismatches, and control delays during operation [Yu et al., 2017]. To address external load disturbances, consensus protocols based on neighbourhood information have been developed to achieve online disturbance estimation and coordinated control of multiple EHAs [China Patent, 2019]. For electro-hydraulic response mismatches during cooperative-mode switching, dynamic torque coordination and fuzzy-control methods have been used to improve hydraulic braking response and switching smoothness [Zhou et al., 2022].

For real-time coordination in complex distributed control systems, dynamic control allocation methods based on robust optimization and linear matrix inequalities have been developed for multi-actuator control allocation, with closed-loop stability being verified [Zhang et al., 2020]. To address network-induced delays, networked sliding-mode control can compensate for time-varying delays and improve the robustness of levelling systems in multi-axle transport vehicles [Yang et al., 2012]. In addition, distributed electro-hydraulic proportional control in high-speed railway track-laying machinery [Li et al., 2016] and leader–follower cooperative control of multiple hydraulic erecting devices [Yong et al., 2023] have demonstrated that appropriate use of local information exchange can improve multi-actuator synchronization and disturbance rejection.

Existing approaches have achieved substantial progress in load-disturbance suppression, synchronized tracking, and compensation for specific communication delays. However, most studies are based on a limited number of nodes, relatively regular disturbances, or predefined communication conditions. For CM DSA characterized by severe impacts, strong nonlinearities, heterogeneous actuators, and dynamically varying communication topologies, the propagation of impact loads across mechanical structures, hydraulic circuits, and control networks, together with the resulting cross-node coupling and synchronization-stability problems, remains insufficiently understood. These common research gaps are further synthesized in [Section 5.4](#).

5.2 Drive, Braking, and Steering Control in DSA

Drive, braking, and steering are the principal control functions governing mobility, safety, and energy management in CM travel systems. The discretization and modularization of drive, braking, and steering actuators in DSA make the system more susceptible to uneven power allocation, wheel slip, and asynchronous actuator responses under variable loads, strong coupling, and impact conditions [Huang et al., 2024].

For drive control, power allocation and slip regulation in distributed electric-drive systems directly affect traction performance and overall SEE. For fuel-cell-based distributed drive systems, hierarchical control, sliding-mode control, and power-signal decomposition have been employed to optimize wheel slip ratio and energy allocation [Li et al., 2024b; Li et al., 2024a; Li et al., 2023]. Under extreme driving conditions, time-varying MPC and distributed sliding-mode control have also been applied to improve path-tracking accuracy and vehicle stability [Shi et al., 2024; Yang et al., 2022].

For braking control, distributed electro-hydraulic braking systems must simultaneously ensure vehicle stability and regenerative energy recovery. Braking-force allocation strategies optimized using genetic algorithms can improve energy-recovery efficiency under dynamic braking conditions [Pei et al., 2020], while sliding-mode-based slip-ratio control can enhance the dynamic response and slip-regulation accuracy of distributed electro-hydraulic brake-by-wire systems [Ji et al., 2023].

For steering control, hydraulic, electric, and steer-by-wire systems have developed along multiple technological routes [Ancha et al., 2007]. For distributed-drive articulated

vehicles, composite nonlinear feedback, active disturbance rejection, and direct-yaw-moment-assisted control have been employed to improve coordinated steering, path tracking, and hydraulic pressure characteristics, respectively [Xu, T. et al., 2024; Cao et al., 2023; Xu et al., 2020].

Existing research remains strongly focused on drive, braking, and steering control within the mobility system, whereas dynamic power coordination among multiple working actuators and between the mobility and working systems has received comparatively limited attention. For DSA with shared electrical sources, independently driven actuators, and energy-regeneration capability, optimizing an individual actuator or a single functional subsystem is no longer sufficient to achieve overall machine-level optimality. Therefore, multi-objective cooperative optimization among task performance, SEE, peak-power constraints, actuator loading, and service life represents an important direction for future system-level control.

5.3 Fault Diagnosis and Fault-Tolerant Control in DSA

CM DSA are characterized by multiple nodes, multi-physical-domain coupling, and harsh operating environments. Consequently, faults in local sensors, actuators, communication links, or power units may affect overall machine coordination. Real-time and accurate fault detection, state estimation, and post-fault control reconfiguration are therefore essential for reliable DSA operation.

Existing studies have investigated these issues at different levels. Finite-time distributed state estimation can reconstruct the global states of nonlinear hydraulic systems through neighborhood information exchange [Zhu et al., 2024]. Blockchain-empowered federated domain generalization can improve machinery fault-diagnosis performance while protecting distributed data [Zhang et al., 2024]. Cloud-edge collaborative transmission, federated learning, and multi-level multi-sensor information fusion also provide new approaches for addressing data silos, distribution shifts, and information redundancy [Wang et al., 2024; Qian et al., 2025; Xiao et al., 2025; Yu et al., 2024; Yu et al., 2026].

For coordination mismatches caused by actuator faults, distributed controllers based on switching control and wireless data transmission can suppress synchronization errors and facilitate rapid post-fault recovery [Dong et al., 2018]. In addition, studies of multi-domain coupled failure modes [Minav et al., 2018] and dynamic-control-allocation-based fault-tolerant methods [Zhang et al., 2018] provide theoretical foundations for distributed fault handling from the perspectives of fault mechanisms and control reconfiguration, respectively.

Most current studies still follow a “local fault–local diagnosis–local reconfiguration” paradigm, whereas faults in DSA may propagate across the actuator physical layer, local control layer, communication layer, and system coordination layer. Degradation of a local actuator not only alters its own closed-loop dynamics but may also affect other nodes through synchronization relationships, power sharing, and task allocation. Future research should therefore extend fault tolerance from individual nodes toward global fault-tolerant mechanisms across physical domains, control layers, and communication layers, while establishing a unified framework that integrates fault propagation, distributed state estimation, control reconfiguration, and resource reallocation.

5.4 Research Gaps and Open Challenges in Distributed Control

The studies reviewed above indicate that DSA control research has established important foundations in multi-actuator synchronization, energy allocation, motion control, and local fault tolerance. Nevertheless, several common problems remain unresolved when extending local control methods toward machine-level distributed cooperative control. As shown in Table 7, it is important to distinguish research gaps in control theory from implementation obstacles associated with onboard CM platforms.

Table 7 Main research gaps in distributed control of DSA.

Research gap	Theoretical challenge	Implementation constraint
Multi-actuator coupling under severe impacts	Impact propagation; nonlinear/multi-timescale stability	Impact sensing; parameter variation; actuator mismatch
Lightweight distributed optimization	Convergence–optimality–complexity trade-off	Limited computation, memory, and bandwidth
Communication-resilient coordination	Stability under delays, packet loss, and topology switching	Jitter, synchronization, bandwidth, and interference
Cross-layer fault tolerance	Fault propagation; distributed estimation and reconfiguration	Limited sensing/data; diagnosis and reconfiguration delays
Heterogeneous multi-timescale coordination	Cross-timescale consistency and stability	Heterogeneous interfaces, sampling rates, and power levels
Performance–energy–lifetime co-optimization	Multi-objective trade-offs; safe intelligent control	Online computation, data quality, and validation

First, nonlinear coupling among multiple actuators under severe impact loads represents a distinctive challenge for CM compared with conventional networked multi-agent systems [He et al., 2025]. Transient impacts generated during digging contact, rapid acceleration/deceleration, and abrupt load variations may propagate among actuation nodes through mechanical structures, hydraulic circuits, and shared power networks, resulting in strongly nonlinear, time-varying, and multi-timescale coupling. Most existing synchronization and consensus-control methods are still based on relatively regular disturbances or low-dimensional coupling models. Mature theoretical frameworks are therefore still lacking for distributed modelling of impact propagation and cross-node dynamics and for guaranteeing global stability under parameter uncertainty and structural variations.

Second, local nodes in DSA may simultaneously undertake state estimation, closed-loop control, neighborhood communication, energy optimization, and even fault diagnosis, whereas onboard embedded controllers are subject to strict constraints on computing capacity, memory, and control-cycle time. Distributed MPC and iterative optimization can explicitly address coupled constraints, but their computational burden increases with the number of nodes, prediction-model complexity, and the number of communication iterations [Christofides et al., 2013]. Lightweight approaches based on model reduction, problem decomposition, explicit or approximate optimization, event-triggered computation, and learning-based surrogates are therefore required to reduce online computation and communication while preserving convergence, stability, and constraint satisfaction. The trade-off among global optimality, convergence speed, and computational complexity remains a fundamental issue in distributed optimization.

Third, the communication network should be treated as an explicit part of the controlled system rather than as an ideal information channel. Practical DSA may simultaneously experience time-varying delays, packet loss, communication jitter, asynchronous sampling, and node entry or departure. Networked-control studies have demonstrated that such effects can alter closed-loop stability and introduce dynamic constraints independent of the mechanical plant [Liu et al., 2019]. Accordingly, further research is required on delay- and packet-loss-tolerant robust or predictive cooperative control, event-triggered communication, and stability analysis under switching topologies, so that coordinated operation can degrade gracefully under non-ideal communication rather than remaining effective only under ideal network conditions.

Fourth, the nodes within DSA are inherently heterogeneous. Electric traction motors, EHAs, EMAs, steering units, and braking units differ substantially in dynamic bandwidth, control period, and power constraints. A multi-layer control system may simultaneously contain millisecond-level local closed loops and slower system-level energy optimization and task coordination. Therefore, multi-timescale coordination among heterogeneous

nodes, together with the relationship between local autonomy and global performance, requires further investigation [Scattolini, 2009].

Fifth, existing control objectives are still frequently centered on tracking error, energy consumption, or a single stability metric, whereas practical CM DSA must simultaneously account for task performance, energy efficiency, thermal loading, peak power, actuator degradation, and lifecycle reliability. Developing multi-objective performance indices with explicit physical constraints and engineering meaning, while coordinating local optimality with global machine-level optimality, is a key step in advancing from cooperative control toward cooperative optimization.

Data-driven methods have already demonstrated potential for extracting and decoupling complex state information in heavy-duty machinery under dynamic loading conditions [Xing et al., 2025]. However, their coverage of training data, generalization under out-of-distribution conditions, and guarantees of closed-loop stability and safety constraints remain insufficient. Future intelligent distributed control should therefore focus not only on prediction or control accuracy, but also on physics-informed constraints, online safety monitoring, and verifiable learning-based control mechanisms.

6 Engineering Challenges and Application Adaptability of DSA in CM

The preceding studies demonstrate that DSA can improve system flexibility, energy efficiency, and fault-isolation capability through modular power supply, independently driven actuators, and node-level coordination. However, CM varies substantially in machine type, power level, and operating environment, and DSA should not be regarded as a universally applicable replacement for existing CSA. Its engineering value must be evaluated comprehensively in relation to machine power level, duty cycle, impact loading, packaging space, cost, and maintenance conditions.

Accordingly, this section goes beyond a simple enumeration of the technical disadvantages of DSA and further examines their practical application boundaries from the perspectives of engineering implementation constraints, computing and communication infrastructure, cybersecurity and inter-node trust, standardization and functional safety, and adaptability to representative machine types.

6.1 System Integration, Cost, Weight, and Lifecycle Constraints

The modular design of DSA improves system-configuration flexibility but simultaneously increases the number of power units, sensors, controllers, and connection interfaces. Stable operation of individual nodes depends on reliable system integration and coordination mechanisms, while an increasing number of modules also increases the number of potential failure points and the complexity of fault diagnosis, maintenance, and spare-parts management [Zhu et al., 2024; Minav et al., 2018]. DSA typically require multiple electric motors, hydraulic pumps, sensors, and control units, resulting in relatively high initial development and manufacturing costs. Highly modular architectures may also increase lifecycle maintenance and management costs [Guo et al., 2022; Minav et al., 2018; Ketelsen et al., 2019].

Independent power units may additionally increase overall machine mass and installed power. Existing studies on distributed hydraulics have shown that local power units, electric motors, and additional structures may increase actuator and working-equipment mass and consequently affect machine motion performance [Niraula et al., 2018]. Meanwhile, satisfying peak-power demands at different nodes may increase total installed power and further complicate energy management [Roth et al., 2019]. The cooperative operation of multiple power sources and modular units changes heat generation from a conventionally centralized distribution toward multiple spatially distributed heat sources. Under high-load and prolonged operation, local electric motors, power electronics, and hydraulic units may experience substantial temperature rise, while

modular packaging can also restrict the design of centralized cooling systems [Maurus et al., 2017; Habibi et al., 2008].

Therefore, engineering evaluation of DSA should not be based solely on instantaneous efficiency or local actuator performance, but should also account for additional mass, installed power, cooling-system requirements, maintainability, and long-term durability. In particular, for heavy-duty CM, long-term reliability data for distributed units under vibration, dust contamination, thermal cycling, and impact loading remain substantially more limited than those available for mature centralized hydraulic systems.

6.2 Onboard Computing, Communication, and Edge–Cloud Collaboration

As the number of DSA nodes increases, relying on a single central controller to simultaneously perform high-frequency sampling, state estimation, cooperative optimization, fault diagnosis, and large-scale data storage creates substantial computational and communication bottlenecks. Conventional cloud-based centralized processing offers considerable computing capability, but communication round-trip latency and network availability prevent it from directly supporting millisecond-level safety-critical closed loops. Conversely, complete reliance on individual local controllers is constrained by embedded computing and storage resources. Therefore, a more suitable information-processing architecture for DSA is likely to involve hierarchical device–edge–cloud collaboration.

Within such an architecture, local actuator controllers are responsible for high-frequency closed-loop control, basic condition monitoring, and safety protection; onboard or site-level edge nodes perform multi-actuator state fusion, short-timescale cooperative optimization, and fault diagnosis; and cloud platforms are more suitable for non-real-time tasks such as model training, historical-data analysis, fleet-level optimization, and predictive maintenance. Industrial IoT studies have shown that Edge–Fog–Cloud and cloud–edge–device hierarchical architectures can provide complementary advantages in computing capability and real-time response [Kumar and Agrawal, 2023; Zhang et al., 2024].

However, edge–cloud collaboration should not be interpreted simply as transferring computational tasks from a central controller to the cloud. Further research is required to determine how tasks should be dynamically allocated among local, edge, and cloud layers, how local safety-oriented degraded operation should be maintained during communication failures, and how edge-computing results can remain temporally consistent with fast physical closed-loop dynamics under CM duty cycles and safety requirements.

6.3 Data Security, Privacy Protection, and Multi-Node Trust

As DSA evolve from physical decentralization toward an information-distributed control layer, extensive data exchange occurs among controllers, sensors, actuators, and external platforms, thereby enlarging the cyberattack surface. Denial-of-service attacks, false-data injection, replay attacks, and unauthorized node access may corrupt state-estimation results and control decisions and, in severe cases, transform information-security threats into physical-safety hazards [Xing and Shen, 2024].

Accordingly, multi-node communication requires comprehensive security mechanisms involving identity authentication, access control, encrypted transmission, node trustworthiness, and anomalous-data detection. On this basis, blockchain represents a potential technology for strengthening trusted multi-node collaboration. Its distributed ledger and tamper-resistant characteristics can support trusted recording of node operation, fault, and maintenance logs and improve data traceability, while smart contracts may further support resource allocation, task scheduling, and energy management at site or fleet level. Existing IoT studies indicate that blockchain can improve data integrity, trust, and traceability in distributed environments [Khan et al.,

2024], while the blockchain-empowered federated fault-diagnosis research already discussed in this review demonstrates its potential for secure machinery-data sharing [Zhang et al., 2024].

Nevertheless, blockchain is not suitable for directly replacing millisecond-level real-time control networks. The computational, communication, storage, and latency overhead associated with consensus mechanisms may conflict with the real-time requirements of CM [Zhang et al., 2024]. More appropriate applications therefore include non-real-time trusted logging, cross-node data auditing, and upper-level resource coordination, while fast closed-loop control should remain within deterministic local control networks.

6.4 Standardization, Interoperability, and Functional Safety

Modularity can deliver its full engineering value only when different modules can be reliably interconnected and replaced. Future DSA may integrate electric motors, EHAs, EMAs, sensors, and local controllers from different suppliers. Without standardized communication protocols, control interfaces, state definitions, and diagnostic-data formats, system-integration and maintenance costs may increase substantially. Studies of distributed industrial cyber-physical systems indicate that interoperability, portable control software, and standardized interfaces are important foundations for scaling DSA from laboratory prototypes to industrial systems [Alves et al., 2025].

Furthermore, increasing the number of nodes, communication links, and software functions also complicates the verification boundaries of safety-related control functions. CM control systems must satisfy requirements related to safe states under faults, redundancy design, and diagnostic coverage. ISO 19014 specifies a functional-safety methodology for safety-related control systems of earth-moving machinery based on Machine Performance Levels [ISO, 2018]. Future DSA therefore need not only demonstrate the stable operation of individual controllers and actuators, but also establish system-level verification and validation methods for the complete DSA, including functional safety under multi-node faults, communication failures, and control reconfiguration.

6.5 Architecture Adaptability for Representative CM Applications

The above challenges indicate that DSA should not be regarded as the universally optimal architecture for all CM. Their practical value depends on machine power level, number of actuators, load spectrum, energy-regeneration potential, and operating environment. Existing studies similarly indicate that different heavy-duty mobile-machine applications impose substantially different requirements on actuators and power-supply architectures and that technological routes should therefore be matched to specific machine types rather than generalized as universally substitutable solutions [Fassbender and Minav, 2020]. Table 8 summarizes the main applicability characteristics and near-term feasible technical routes of the two architectures for small/compact excavators, wheel loaders, large excavating/lifting equipment, and large-tonnage mining machinery.

Table 8 Applicability of DSA and CSA in Representative CM Scenarios.

Application Scenario	DSA Suitability	Near-Term Architecture	CSA Advantages
Small/compact excavators	High	EHA, zonal hydraulics, highly DSAs	Low cost; simple architecture
Wheel loaders	High, impact-constrained	Distributed electro-hydraulic or hybrid architectures	High impact capability; peak-power handling
Large excavating/lifting equipment	Moderate; selective decentralization	Key-actuator EHA + centralized hydraulic backbone	High power density; mature integration
Large-tonnage mining equipment	High for mobility; limited for working implements	Distributed electric traction + centralized/hybrid hydraulics	Continuous high power; robustness; maintainability

Small and compact CMs typically have low-to-medium power demand, a limited number of actuators, and comparatively moderate peak-power requirements. Urban construction also places strong emphasis on low noise, zero local emissions, and high energy efficiency. These characteristics make such machines particularly suitable for early adoption of distributed hydraulics and fully electric actuation technologies. Existing studies on decentralized hydraulics for micro-excavators have demonstrated the energy-saving potential of this approach [Zhang et al., 2017b], while reviews of P-ECM likewise show that compact equipment represents one of the more active categories in current electrification development [Huang et al., 2024; Lin et al., 2020].

Wheel loaders operate with highly repetitive short duty cycles and provide substantial opportunities for potential-energy recovery; therefore, both distributed electric traction and distributed implement actuation have considerable potential [Fei et al., 2023; Gao et al., 2022]. However, the digging phase also involves severe impact loading and high peak-power demand, meaning that completely eliminating the centralized hydraulic system may not yet provide an overall engineering advantage. Hybrid centralized–distributed solutions that arrange EHA/EMA units in parallel with conventional hydraulic systems can provide a compromise among energy-saving benefits, installation space, and rated-power requirements [Mu et al., 2025], and may therefore represent an important transitional route toward highly DSAs for wheel loaders and medium-to-large excavators.

For large-tonnage mining machinery, distributed technologies such as wheel-side electric drives already have a relatively clear technical basis in the mobility system, whereas working implements typically operate under continuously high power and severe impact loading. The high-power density, overload capability, and mature maintenance infrastructure of hydraulic systems therefore continue to provide important advantages in such applications [Quan et al., 2021; Fassbender et al., 2021]. Consequently, large machines are more likely to adopt partial functional decentralization or hybrid centralized–distributed architectures before transitioning toward fully independent power supply for all actuators.

In summary, CSAs and DSAs should not be regarded simply as mutually substitutable technologies; instead, they exhibit clear application-dependent boundaries. DSA are more suitable for applications in which independent actuator power supply, energy regeneration, modularity, and digitalization can provide substantial system-level benefits. In contrast, mature CSA retain important, and in some cases difficult-to-replace, engineering advantages in applications dominated by extremely high power, severe impact loading, stringent reliability requirements, and strong cost and maintenance constraints. Future CM is therefore more likely to exhibit long-term coexistence of centralized, hybrid, and highly distributed architectures, with the degree of decentralization optimized according to specific machine types and operational tasks.

7 Conclusion and Outlook

This paper systematically reviews the concepts, architectural evolution, drive units, control methods, and engineering challenges of DSA for CM. The main conclusions are as follows:

(1) DSA in CM are not independent of CSAs, but represent a continuous technological spectrum driven jointly by power-source electrification, independent actuator power supply, and modular actuation. Their essential evolution is from centralized power supply toward actuator-level independence, thereby improving SEE and flexibility by shortening energy-transmission paths, reducing throttling losses, and mitigating multi-actuator coupling. Centralized, hybrid centralized–distributed, and highly distributed architectures are therefore expected to coexist across different power levels and operating conditions.

(2) EHA and EMA are currently the two principal distributed drive-unit technologies, but they exhibit different application boundaries. Owing to their high power/force density and stronger adaptability to heavy loads and impacts, EHA currently have greater

engineering maturity for heavy-duty CM. EMA offer shorter energy-conversion chains, high control accuracy, and leakage-free operation, but their impact adaptability, thermal management, mechanical-transmission durability, and long-term reliability under heavy-duty conditions remain insufficiently validated. The two technologies should therefore be regarded as complementary solutions for different power levels and duty conditions rather than as direct substitutes.

(3) DSA control is evolving from local node-level regulation toward multi-node cooperative optimization. Existing studies have addressed synchronization control, coordinated drive–braking–steering control, and fault diagnosis and tolerance. However, nonlinear multi-actuator coupling under severe impacts, lightweight distributed optimization under resource constraints, cooperative stability under non-ideal communication, heterogeneous multi-timescale coordination, and cross-layer global fault tolerance remain major research gaps.

(4) The engineering value of DSA is strongly application-dependent. Small and compact machines can more readily benefit from modularization, electrification, and independent power supply. Cyclic machines such as wheel loaders also offer substantial potential, although severe impacts and high peak-power demands make hybrid architectures attractive in the near term. For large and high-tonnage machines, centralized or hybrid hydraulic systems remain highly competitive because of their power density, overload capability, reliability, and mature maintenance infrastructure. Meanwhile, as DSA become increasingly digitalized and connected, onboard computing, cybersecurity, data privacy, multi-node trust, and functional safety are emerging as additional engineering constraints.

Future research can be organized into the following three levels:

(1) Theoretical research. Future work should establish distributed dynamic modelling and control frameworks for multi-actuator systems subject to severe impacts, strong nonlinearities, and multi-physical-domain coupling. Particular attention should be given to cooperative stability under communication delays, packet loss, and dynamic topologies, together with cross-layer fault propagation and global fault-tolerant mechanisms. Lightweight distributed optimization should also be developed to balance real-time performance, stability, and constraint satisfaction, while extending control objectives toward joint optimization of task performance, energy efficiency, thermal loading, and component lifetime.

(2) Component-level development. For EHA, priority should be given to thermal management, sealing reliability, lightweight design, and long-term durability under highly integrated configurations. For EMA, further improvements are required in motor and mechanical-transmission power/force density, impact-load capability, thermal performance, and lifecycle reliability. In parallel, modular and standardized interfaces for drive units, sensors, and local controllers should be developed to facilitate rapid integration and interchangeability among heterogeneous actuator technologies.

(3) Industrial application and deployment. Research should progress from proof-of-concept prototypes toward long-term full-machine validation under representative duty cycles, with centralized, hybrid, or highly distributed architectures selected according to machine type, power level, and load characteristics. For digital infrastructure, a hierarchical local-control–edge-coordination–cloud-analysis architecture can balance real-time requirements and computational resources, while authentication, encrypted communication, anomaly detection, and multi-node trust mechanisms should be incorporated. Blockchain is more appropriately used for trusted logging, data auditing, and upper-level task or resource coordination rather than real-time closed-loop control. Further development of interoperability standards, functional safety, and system-level verification frameworks will be essential for translating DSA from prototype research into large-scale industrial applications.

Declaration of competing interest

The authors declare no known competing financial interests or personal relationships that could have influenced the work reported in this paper. AI-assisted tools were used only for language polishing and formatting checks, and the authors take full responsibility for the content.

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